MSci Individual Project

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Statistical Tests to Detect the Number of Clusters Present in the Data

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April 14, 2026

This is my own work except where otherwise stated.

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Abstract

Clustering is an important tool in data analysis however a fundamental question is how many clusters are truly present in the data, as opposed to clusters simply due to sampling variation. In this project we consider methods that attempt to answer this question.

We follow the path of the SigClust method (which tests whether or not to split data) from its introduction by Liu et al. (2008), to the hierarchical variant proposed by Kimes et al. (2017) which can be used to predict the number of clusters present in data. We also explore another method, called RIFT, proposed by Chakravarti et al. (2019) which similarly has variants that allow it to determine the number of clusters present in data, we suggest a modification to their L_2 method.

We conclude with a thorough simulation study to compare the methods presented within the report, our results are consistent to those of Chakravarti et al. (2019) and Kimes et al. (2017), however they add reliability due to the new scenarios tested. Though, we find unlike the original L_2 variant, the modified L_2 method performs similarly to the other methods. Finally we apply these methods to a breast cancer gene dataset and find that the RIFT methods perform well.

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Chapter 1

Introduction

Clustering is the task of partitioning a dataset into disjoint groups. This is done such that the observations within each group are quite similar to each other, while observations in different groups are quite different from each other (James et al., 2021).

Clustering has many applications from finance (Gupta and Huefner, 1972) to genomics (Eisen et al., 1998) to climatology (Huth et al., 2008), and more. There exists many clustering algorithms to obtain a clustering from data (see Xu and Tian (2015) for a comprehensive survey), however a number of these algorithms require the number of clusters of interest to be pre-specified. Specifying this number is no less important than the clustering algorithms themselves, as a misspecification could lead to the resulting clusters being quite different from the true clusters, and therefore it could be potentially misleading to any following analysis.

Hence there exists a need to determine the true number of clusters present in data. The challenge in deciding the number of clusters comes from differentiating between clustering due to sampling variation and clustering due to the underlying structure of the data. To overcome this there are methods that attempt to determine the number of statistically significant clusters present, of which some are explored in-depth in this report.

One such method is the hierarchical SigClust procedure proposed by Kimes et al. (2017) which, assuming Gaussian clusters, attempts to determine the number of statistically significant splits in a dataset. This method is based off of the SigClust method first proposed by Liu et al. (2008) and later extended by Huang et al. (2015). The extensions by Huang et al. (2015) relate to the estimation of the covariance matrix of the cluster, as one of the issues pertaining to the original method was that it was often either too conservative or too anti-conservative. Using these extensions Kimes et al. (2017) proposed the hierarchical version, however a disadvantage of this method, inherent from the SigClust procedure, is the assumption of Gaussian clusters.

An alternative approach by Chakravarti et al. (2019), RIFT, relaxes the assumption of Gaussianity of clusters required and instead tests *relative* fit of Gaussian clusters to a dataset. From a practical standpoint this relative fit is far more desirable as in most circumstances these models are regarded as approximations. Like the Hierarchical SigClust procedure, Chakravarti et al. (2019) apply their method hierarchically to allow for determination of the number of statistically significant clusters. However Chakravarti et al. (2019) also propose a sequential version of the RIFT method.

The SigClust method and its sequential variant have been analysed using a variety of simulations (Liu et al., 2008; Kimes et al., 2017). Likewise, Chakravarti et al. (2019) has also explored the RIFT method in various simulations as well as comparing it to other methods. Despite this generally simulation studies in the multiple cluster cases are few compared to the base case (testing only if data is from one cluster).

In this report we first detail the SigClust, RIFT, AIC and BIC methods to determine the number of clusters present in a dataset. Our contributions are the proposal of a modification to the importance sampling procedure used in sequential RIFT which results in the L_2 and Kullback–Leibler divergence variants preforming similarly contrary to without this modification. All methods are also implemented in R and an extensive simulation study comparing them is performed in cases not considered by Chakravarti et al. (2019) and Kimes et al. (2017), as well as with more methods. Following this, we apply these methods to a BRCA (breast cancer gene) dataset which has not been done for the RIFT methods.

The remainder of this report is structured as follows. Chapter 2 provides background definitions required for the clustering methods which are presented and defined in chapter 3. Chapter 4 explains in depth the AIC, BIC, Sigclust and RIFT methods including variants. While chapter 5 details the simulation study performed and summarises the comparisons between the methods in several scenarios. The methods are then applied to real life data in chapter 6. And finally chapter 7, the conclusion, includes a summary of the results as well as a discussion of possible future work.

Chapter 2

Cluster Definitions

The seemingly simple question of what a cluster is is perhaps more contentious than one would assume. The definitive notion of a cluster is not rigorously agreed upon among all mathematicians (and non mathematicians), which ultimately leads to varying clustering algorithms and corresponding analysis. This disaccord is explored further in (Estivill-Castro, 2002). Most commonly, and perhaps intuitively, items within a cluster are "similar", while items from different clusters are "different". However, this requires us to ask the question of how to measure similarity. In this chapter I explore different definitions of a cluster, and their corresponding similarity measures.

There are many different ways of clustering. For a comprehensive list one can see (Xu and Tian, 2015). Some of the most common include:

- Hierarchical clustering
- Partition based clustering
- Distribution based clustering
- Density based clustering
- Graph theory based clustering
- Fuzzy clustering

In this paper I focus on certain hierarchical, partition based (more specifically centroid based), and distribution based clustering methods.

2.1 Hierarchical Clustering

Hierarchical clustering, (Kumar, 2020), forms a hierarchy among the data, and clusters are defined by splitting the data at certain points in this hierarchy. Broadly speaking hierarchical clustering can be divided into two categories:

- Agglomerative - Bottom up clustering, where each observation begins in its own cluster and the two clusters nearest to each other, based off a linkage criterion, are merged. Where the linkage criterion is a metric that defines the notion of similarity measure between clusters. This is repeated until all observations are assigned to the same cluster.
- Divisive - Top down clustering, all observations begin in the same cluster and splits are performed on the cluster based off certain criterion. There are many valid criteria, (Savaresi et al., 2002), however common criteria include: splitting every cluster thus creating a complete binary tree, splitting the cluster with the largest number of elements or splitting the

cluster with the largest variance. The split is often performed to minimise the within cluster dissimilarity of the two new clusters. This is done until all observations belong to individual clusters.

A clustering is then defined by cutting the hierarchy, with points still connected after the cut being in the same cluster and the height of the cut determining the number of clusters defined. This cut point is discussed further below and in chapter 4.

It is important to note that hierarchical clustering only produces clusters that are embedded within each other.

2.1.1 Dissimilarity Metric

To allow us to define a linkage criterion we first need to define a dissimilarity metric, d, between data points. A notion of dissimilarity is essential for hierarchical clustering, and different choices of metrics can produce different shaped clusters (Nielsen, 2016). The most common metric is the Euclidean distance:

$$\begin{gathered} d(x,y) := \|x-y\| = \sqrt{\sum_{i=1}^{n}(x_i-y_i)^2} \\ \text{where } x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n, y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n \end{gathered} \tag{2.1}$$

An example whereby different metrics produce different results is the case $a = (0,0)$, $b = (3,3)$, $c = (4,0)$:

- The Euclidean distance gives $\|a-b\| = \sqrt{18} \approx 4.2$, $\|a-c\| = 4$ and so c is closer to a than b.
- Using the Chebychev distance ($\|x-y\|_\infty := \max_{i=1,\dots,n} |x_i - y_i|$), $\|a-b\|_\infty = 3$, $\|a-c\|_\infty = 4$ and hence b is closer to a than c.

Hence selecting an appropriate metric in context of the problem is of utmost importance.

2.1.2 Linkage Criterion

The linkage criterion defines the notion of similarity between clusters (sets of observations). Consider two clusters $\mathcal{A}$ and $\mathcal{B}$ then some common linkage criteria are:

- Complete linkage: $\max\{d(a,b) \text{ s.t. } a \in \mathcal{A}, b \in \mathcal{B}\}$ where d is a dissimilarity metric .
- Single linkage: $\min\{d(a,a) \text{ s.t. } a \in \mathcal{A}, b \in \mathcal{B}\}$ where d is a dissimilarity metric .
- Wards linkage which is defined below.

Ward's Linkage

In his seminal paper, (Ward Jr, 1963), proposed an objective function, which is commonly referred to as Ward's linkage, as a means of selecting the optimal clusters to divide. It is based on the error sum of squares also called cluster sum of squares, which is defined as follows, for a set of data points $\mathcal{A}$:

$$ESS_{\mathcal{A}} := \sum_{a \in \mathcal{A}} \left\|a - \bar{\mathcal{A}}\right\|^2$$

Where for a collection of data points $\mathcal{A}$, I define $\bar{\mathcal{A}} := \frac{1}{|\mathcal{A}|}\sum_{a\in\mathcal{A}} a$, the mean of all elements of $\mathcal{A}$. Intuitively, notice $\frac{ESS_{\mathcal{A}}}{|\mathcal{A}|-1}$ is exactly the sample variance of $\mathcal{A}$, and so $ESS_{\mathcal{A}}$ is proportional to the

sample variance of $\mathcal{A}$. Additionally, $ESS_{\mathcal{A}}$ can be thought of as the information loss that results from treating all individual data points in $\mathcal{A}$ by their mean $\bar{\mathcal{A}}$ (Ward Jr, 1963).

This can be extended to evaluate a grouping of the data $\mathcal{A}$. That is given disjoint clusters $C_1, ...C_k$, that partition $\mathcal{A}$, we can define the error sum of squares of this grouping as: $ESS_{\{C_1,...,C_k\}} := \sum_{i=1}^{k} ESS_{C_i}$. This in turn can be used to compare two groupings of data, consider two partitions of a given data set: $C_1, ..., C_k$; $\tilde{C}_1, ..., \tilde{C}_{\tilde{k}}$, these can be compared by finding the difference of their respective ESS's:

$$ESS_{\{C_1,...,C_k\}} - ESS_{\{\tilde{C}_1,...,\tilde{C}_{\tilde{k}}\}} \tag{2.2}$$

(2.2) is the objective function Ward proposed (Ward Jr, 1963) and is often called Ward's linkage.

Intuitively, (2.2) is (proportional to) the increase in (sample) variance using clustering $\{C_1, ..., C_k\}$ rather than clustering $\{\tilde{C}_1, ..., \tilde{C}_{\tilde{k}}\}$. Or in terms of information loss, it is the information lost by using the clustering $\{C_1, ..., C_k\}$ rather than $\{\tilde{C}_1, ..., \tilde{C}_{\tilde{k}}\}$.

Consider equation (2.2) applied hierarchically, whereby initially the data is partitioned into k clusters, $C_1, C_2, ..., C_k$, and without loss of generality clusters C_1, C_2 are combined to form $\tilde{C}_1$, allowing the simplification of the equation (2.2):

$$\begin{aligned}(2.2) &= ESS_{\{\tilde{C}_1,C_3,C_4,...,C_k\}} - ESS_{\{C_1,...,C_k\}} \\ &= ESS_{\tilde{C}_1} - [ESS_{C_1} + ESS_{C_2}]\end{aligned}$$

And so the information loss/variance gain in the hierarchical setting is simply the difference between the ESS of the combined cluster and the uncombined clusters.

2.1.3 Dendrograms

Hierarchical clustering results are generally presented using a dendrogram. A dendrogram shows the hierarchical relationship between clusters, as seen in figure 2.1. The height at which two objects are joined corresponds to their similarity, ie individuals joined only high up in the dendrogram are relatively dissimilar, whereas those linked near the bottom of the dendrogram are relatively similar. Consider the example shown in figure 2.1. Whereby the highest split separates the data to $\{x_1, x_2, x_3, x_4, x_5\}, \{x_6, x_7\}$, (this corresponds to the split at the blue circle), since this is the highest split in the dendrogram this signifies that these two clusters are most dissimilar. Conversely the final split (marked by a green circle) separates x_6 and x_7 indicating these two points are most similar.

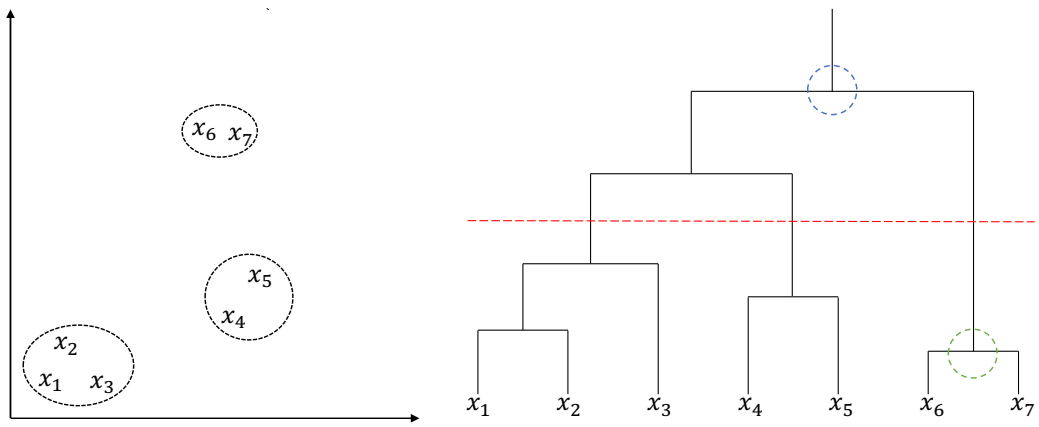


Figure 2.1: Seven data points $(x_1, ..., x_7)$ and their corresponding dendrogram. With the clusters marked on the left plot corresponding to cutting the dendrogram by the red dashed line.

As their names imply, the bottom up method forms the dendrogram from the bottom up, while the top down method forms the dendrogram from the top down. Once the dendrogram has been formed clusters are found by drawing a horizontal line through the dendrogram (indicated by red dashed line in figure 2.1), and observations that are joined below this line are in the same cluster, with the height of the line determining the number of clusters. In figure 2.1, the red dashed line has partitioned the data into 3 clusters: $\{x_1, x_2, x_3\}, \{x_4, x_5\}, \{x_6, x_7\}$. The height of this line has been taken rather arbitrarily, and deciding this (or equivalently the number of clusters of the data) often poses a significant problem, which is discussed further in chapter 4.

2.2 Centroid-based Clustering

Centroid based clustering is where each cluster is defined by a centroid point, which need not be within the data set. Points are assigned to a cluster based on which centroid they are the nearest to using a dissimilarity metric. Most commonly (as in k-means clustering) the squared Euclidean distance (2.1) is the distance metric used.

Let $\{C_1, ..., C_k\}$ denote k nonempty clusters (disjoint partitions of the data), and d a distance metric. Then given k centroids $\left\{C_1^{(c)}, ..., C_k^{(c)}\right\}$, the corresponding corresponding centroid based clustering is:

$$\underset{\{C_1, C_2, ..., C_k\}}{\operatorname{argmin}} \sum_{j=1}^{k} \sum_{x \in C_j} d\left(x, C_j^{(c)}\right) \tag{2.3}$$

The well known k-means algorithm is an example of a centroid based clustering algorithm (which optimises (2.3) over the centroids). This is discussed in detail in chapter 3.

2.3 Distribution-based Clustering

Another definition of a cluster is data that has been drawn from the same distribution. This can be restrictive since it assumes the underlying distribution of the clusters, however it does allow complex models for clusters that can account for correlation and dependence between attributes. It should be noted that true clusters might not conform to a known mathematical distribution, and so the necessary prior specification of the distribution(s) can also limit this clustering method.

2.3.1 (General) Finite Mixture Models

Assuming distribution based clustering is the equivalent of assuming that the underlying population has a mixture distribution, and the probabilistic model corresponds to a mixture model, (Picard, 2007). Given a population of finite k clusters, where each cluster corresponds to a distribution with probability density function (pdf) f_j for $j = 1, ..., k$, let π_j be the relative size of each cluster, its mixing weight, (where $\pi_j > 0, j = 1, ..., k$ and $\sum_{j=1}^{k} \pi_j = 1$).

The pdf of the population, denoted f, is a mixture distribution of the k densities $f_1, ..., f_k$, and has the form:

$$f(x_i; \theta) = \sum_{j=1}^{k} \underbrace{\pi_j}_{\text{mixing weight}} \cdot \underbrace{f_j(x_i; \theta_j)}_{\text{cluster distribution}} \tag{2.4}$$

The pdf of the jth cluster f_j is called a component density of the mixture model, while π_j is its corresponding weight and $k \in \mathbb{N}$ is the number of mixture components. θ, θ_j correspond to unknown parameters of the pdfs.

The restrictions on the π_j's ($\pi_j > 0, j = 1, ..., k$ and $\sum_{j=1}^{k} \pi_j = 1$) ensure f is a valid probability density function (non-negative and integrates to 1):

$$\pi_j > 0, f_j(x_i; \theta_j) \geq 0 \implies \sum_{j=1}^{k} \pi_j f_j(x_i; \theta_j) = f(x_i; \theta) > 0$$

$$\int_{-\infty}^{\infty} f(x; \theta) dx = \int_{-\infty}^{\infty} \sum_{j=1}^{k} \pi_j f_j(x; \theta_j) dx = \sum_{j=1}^{k} \pi_j \int_{-\infty}^{\infty} f_j(x; \theta_j) dx = \sum_{j=1}^{k} \pi_j = 1$$

Let $X_1, ..., X_n$ be a random sample of size n. Associated with each X_i is the unobserved label Z_i indicating the cluster (called the component when pertaining to mixture models), that is, $Z_i = z_i$ means X_i comes from component z_i. Each Z_i is assumed to be independent and drawn from a generalised Bernoulli distribution, with probabilities $(\pi_1, ..., \pi_n)$. That is:

$$Pr(Z_i = z_i) = \begin{cases} \pi_j & \text{if } z_i = j, \text{ for } j = 1, ..., k \\ 0 & otherwise \end{cases}$$
$$\text{where } \pi_i \geq 0, \sum_{i=1}^{k} \pi_i = 1 \tag{2.5}$$

hence sampling from mixture model can be thought of as two steps:

Step 1: Sample identically and independently distributed Z_i from probability mass function as in (2.5).

Step 2: Sample X_i from the Z_ith mixture component (f_{Z_i}).

The mixture components are usually not arbitrary, but instead belong to a parametric family. Of particular interest (for the RIFT and SigClust methods discussed in chapter 4) is the specific case when all components are Gaussian. This is called a Gaussian mixture model. We discuss estimating the parameters of mixture distributions in chapter 3.

2.3.2 Gaussian Mixture Models

A Gaussian mixture model is a specific form of a mixture model whereby all components f_j are (multivariate) Gaussian distributions. Therefore the general equation for a mixture model (2.4) becomes equation (2.6):

$$f(x; \theta) = \sum_{j=1}^{k} \pi_j \cdot \underbrace{\det(2\pi\Sigma_j)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x - \mu_j)^\top \Sigma_j^{-1}(x - \mu_j)\right)}_{\text{pdf of } \mathcal{N}(\mu_j, \Sigma_j)} \tag{2.6}$$

where μ_j, Σ_j correspond to the mean vector and covariance matrix for the j-th component. To illustrate the flexibility of Gaussian mixture models consider figure 2.2, whereby only two mixture components have produced a density that is not like either of its constituent densities.

Remark. Gaussian mixture models are themselves generally not Gaussian, and this does not contradict the well known fact that a linear combination of independent Gaussian random variables is normally distributed. To see this consider how one would sample from each of these scenarios. Let $X_1 \sim \mathcal{N}\left(\mu_{X_1}, \sigma^2_{X_1}\right)$, $X_2 \sim \mathcal{N}\left(\mu_{X_2}, \sigma^2_{X_2}\right)$ independently, with pdf's f_{X_1}, f_{X_2} respectively.

- To sample from a Gaussian mixture model: Consider sampling from mixture model with component densities f_{X_1}, f_{X_2} and component weights $\frac{1}{2}$, that is sampling Y_1 with pdf:

  $$f_{Y_1}(y) = \lambda f_{X_1}(y) + (1 - \lambda) f_{X_2}(y)$$

  for $\lambda \in (0, 1)$, this is done by:

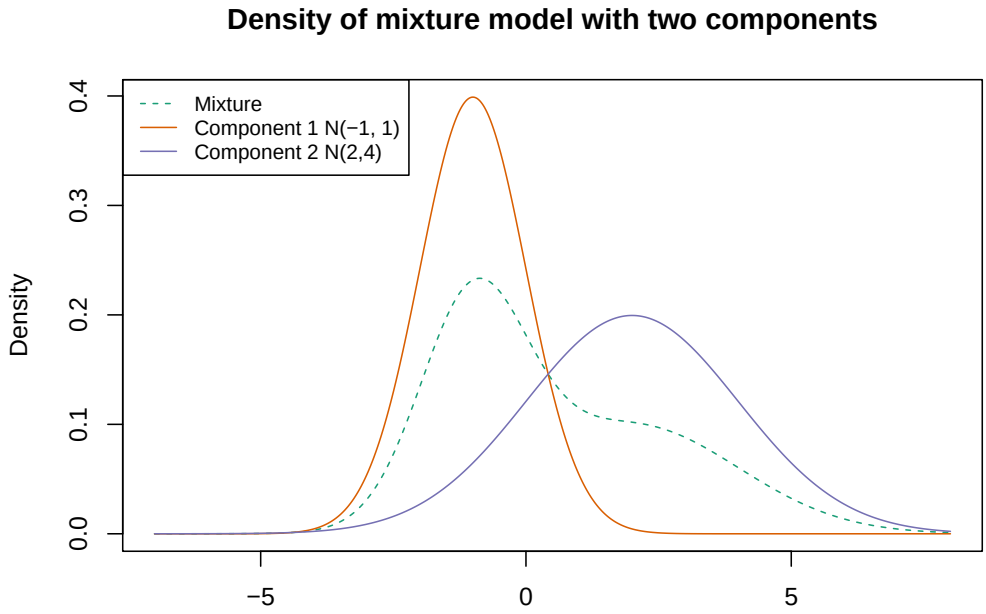


Figure 2.2: Gaussian mixture density with components: $\mathcal{N}(-1, 1)$, $\mathcal{N}(2, 4)$ and respective weights $0.5, 0.5$.

Step 1: Sample z from Z, a generalised Bernoulli distribution (with probabilities $\lambda, 1 - \lambda$)

$$Pr(Z = z) = \begin{cases} \frac{1}{2} & \text{if } z = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

to obtain the mixture component to sample from.

Step 2: Sample from X_z.

- To sample from a linear combination of Gaussian random variables. Consider sampling from Y_2 where $Y_2 := X_1 + X_2$.
  1. Sample from X_1 and X_2
  2. Sum the samples.

These two sampling methods are clearly different and highlight that a Gaussian mixture model is different from a linear combination of Gaussian distributions.

This definition of a cluster- that is data in the same cluster are drawn from the same Gaussian distribution is used extensively throughout this report. However, it should be noted that according to this definition data from other distributions ie. Cauchy, uniform or t-distributions do not correspond to a cluster, and so it is indeed a strong assumption. This definition has been very widely used in literature (Thorsson et al., 2018; Chandriani et al., 2009).

Chapter 3

Clustering Algorithms

Due to the ambiguity of what a cluster is as discussed in chapter 2 there is a wide range of corresponding clustering algorithms. However, I restrict analysis to the k-means algorithm and the EM algorithm for fitting Gaussian mixture models.

3.1 K-means Clustering

The k-means clustering algorithm aims to partition n observations into k centroid based clusters. As discussed in chapter 2 the objective in centroid based clustering is to find the clustering that minimises the total within cluster distance, ie to solve (2.3). In k-means the metric used is the squared Euclidean distance, that is: $d(x, y) := \|x - y\|^2$ and the centroids are defined to be the mean of all points in the corresponding cluster ($C_j^{(c)} = \bar{C}_j$). Hence (2.3) reduces to (3.1).

$$\underset{\{C_1, C_2, \ldots, C_k\}}{\operatorname{argmin}} \left\{ \sum_{j=1}^{k} \sum_{x \in C_j} \left\| x - \bar{C}_j \right\|^2 \right\} \tag{3.1}$$

To see that $\bar{C}_j$ is indeed the optimal choice for $C_j^{(c)}$ when using the squared Euclidean distance consider minimising the objective function in equation (2.3) with respect to the centroids. To do this begin by differentiating with respect to $C_m^{(c)}$, for arbitrary index m, and setting to 0:

$$\begin{aligned}
\frac{\partial}{\partial C_m^{(c)}} \sum_{j=1}^{k} \sum_{x \in C_j} \left\| x - C_j^{(c)} \right\|^2 &= \sum_{x \in C_m} \frac{\partial}{\partial C_m^{(c)}} \left(x - C_m^{(c)} \right)^\top \left(x - C_m^{(c)} \right) \\
&= \sum_{x \in C_m} \frac{\partial}{\partial C_m^{(c)}} \left(x^\top x - 2x^\top C_m^{(c)} + \left[C_m^{(c)} \right]^\top C_m^{(c)} \right) \\
&= \sum_{x \in C_m} -2x + 2C_m^{(c)}
\end{aligned}$$

$$\frac{\partial}{\partial C_m^{(c)}} \sum_{j=1}^{k} \sum_{x \in C_j} \left\| x - C_j^{(c)} \right\|^2 = 0 \implies C_m^{(c)} = \frac{1}{|C_m|} \sum_{x \in C_m} x = \bar{C}_m$$

This is indeed a minimum as the hessian (which is $\operatorname{diag}(2|C_1|, \ldots, 2|C_k|)$) is clearly always positive definite. To solve (3.1) naively one could evaluate the objective function at all possible partitions of the data, however this is usually intractable. For a data set of n points and k clusters there are $S(n, k)$ possible partitions (Celebi et al., 2013) (where $S(n, k)$ denotes the Stirling numbers of the second kind). $S(n, k)$ becomes intractably large even for relatively conservative n and k values,

for example, $S(n = 100, k = 5) \approx 10^{67}$. And hence evaluating (3.1) at each possible partition to find the minimum is intractable except for very small n. In particular this problem can be proven to be NP hard (Aloise et al., 2009).

There exists algorithms that attempt to find the optimum of (3.1), the simplest being Lloyd's algorithm:

Step 1: Initialise: $C_j^{(c)(1)}, j = 1, 2, \ldots, k$ at random unique data points, where $C_j^{(c)(t)}(1)$ denotes centroid j at iteration t.

Step 2: Assign each data point to the nearest centroid (using squared Euclidean distance), where $C_j^{(t)}$ is the cluster defined by all points closest to centroid j (at iteration t).

Step 3: Update the centroids of the clusters:

$$C_j^{(c)(t+1)} = \bar{C}_j^{(t)}$$

Steps 2, 3 are repeated until convergence (or a maximum number of iterations has been reached). It should be noted that Lloyd's algorithm is very sensitive to the initial centroids and often it is repeated several times and the best partition is chosen. This is the 'naive' implementation of k-means, and much faster (more complex) algorithms exist (Pelleg and Moore, 1999) which attempt to find the optimum of (3.1).

3.2 Mixture Models Using the EM Algorithm

The Expectation Maximisation (EM) algorithm (Dempster et al., 1977) is an iterative procedure which attempts to find the maximum likelihood estimate of a set of parameters. It can be used to fit mixture models to data. Below I sketch the derivation and method of the EM algorithm for fitting a mixture model. However a more detailed derivation can be found in Dempster et al. (1977); McLachlan and Krishnan (2007).

Assume we have obtained a random sample and wish to fit a mixture model of the form (2.4). Using the notation as in chapter 2, where θ_j are the unknown parameters associated with the component density f_j, and $\theta = (\pi_1, ..., \pi_{k-1}, \theta_1, ..., \theta_k)$ are all the unknown parameters for f (note the restriction $\sum_{j=1}^{k} \pi_j = 1$, means that π_k is known given $\pi_1, ..., \pi_{k-1}$).

The optimal model can be found by maximising the likelihood function. Recall the likelihood function has the form:

$$\begin{aligned} L(\theta) &= \prod_{i=1}^{n} f(x_i; \theta) \\ \ell(\theta) &= \sum_{i=1}^{n} \log f(x_i; \theta) \\ &= \sum_{i=1}^{n} \log \left(\sum_{j=1}^{k} \pi_j f_j(x_i; \theta_j) \right) \end{aligned}$$

Continuing the sketch of the derivation of the EM algorithm it is natural to use "probability" notation, as in Wasserman (2006), that is $Pr()$, since this allows a more intuitive derivation. Where $Pr(Y = y)$ is defined in the usual way, ie for continuous (discrete) random variable Y,

$Pr(Y = y) = f_Y(y)$ where f_Y is the corresponding pdf (pmf).

$$\begin{aligned}
\ell(\theta) &= \sum_{i=1}^{n} \log\left(\sum_{j=1}^{k} Pr(Z_i = j, X_i = x_i; \theta)\right) \\
&= \sum_{i=1}^{n} \log\left(\sum_{j=1}^{k} \frac{\gamma_{ij}}{\gamma_{ij}} Pr(Z_i = j, X_i = x_i; \theta)\right) \\
&= \sum_{i=1}^{n} \log\left(\sum_{j=1}^{k} \gamma_{ij} \frac{Pr(Z_i = j, X_i = x_i; \theta)}{\gamma_{ij}}\right) \qquad (3.2)
\end{aligned}$$

Where $\gamma_{ij} := P\left(Z_i = j; x_i, \theta\right)$ (Rudin, 2021). The form of (3.2), a log of sums, is difficult to work with, motivating it's simplification through Jensen's inequality. As γ_{ij} is a probability and $-\log()$ is a convex function Jenson's inequality (Durrett, 2019) can be applied to (3.2) to give a lower bound for the likelihood function:

$$-\log\left(\sum_{j=1}^{k} \gamma_{ij}^t \frac{P(Z_i = j, X_i = x_i; \theta)}{\gamma_{ij}^t}\right) \leq -\sum_{j=1}^{k} \gamma_{ij}^t \log\left(\frac{P(Z_i = j, X_i = x_i; \theta)}{\gamma_{ij}^t}\right)$$

With $\gamma_{ij}^t := P\left(Z_i = j; x_i, \theta^t\right)$ analogous to γ_{ij}, except the index t has been added to refer to the parameters γ_{ij} is evaluated at, this notation becomes useful when considering the iterative procedure.

$$\begin{aligned}
\sum_{i=1}^{n} \log\left(\sum_{j=1}^{k} \gamma_{ij}^t \frac{P(Z_i = j, X_i = x_i; \theta)}{\gamma_{ij}^t}\right) &\geq \sum_{i=1}^{n}\sum_{j=1}^{k} \gamma_{ij}^t \log\left(\frac{P(Z_i = j, X_i = x_i; \theta)}{\gamma_{ij}^t}\right) \\
\implies A(\theta, \theta^t) := \sum_{i=1}^{n}\sum_{j=1}^{k} \gamma_{ij}^t \log\left(\frac{P(Z_i = j, X_i = x_i; \theta)}{\gamma_{ij}^t}\right) &\leq \ell(\theta)
\end{aligned}$$

The auxiliary function $A(\cdot, \theta^t)$ provides a lower bound to the likelihood. It can be shown that the maximum of A corresponds to a (local) maximum of ℓ (McLachlan and Krishnan, 2007). And so working with A is preferred since it is easier to maximise than ℓ.

The iterative procedure is therefore (where θ^t denotes the parameter values at iteration t) (Rudin, 2021):

Step 1: 'Guess' a feasible value of $\theta^0 = (\pi_1, ..., \pi_{k-1}, \theta_1, ..., \theta_k)$

Step 2: Until there is little change in θ ($\theta^t \approx \theta^{t-1}$) repeat:

(a) (E-Step): Compute γ_{ij}^t using the most recently computed parameters θ^t.

(b) (M-Step): Using γ_{ij}^t (as computed above) calculate: $\theta^{t+1} = \text{argmax}_\theta\, A(\theta, \theta^t)$.

The γ_{ij}^t term can be calculated using Bayes rule:

$$\gamma_{ij}^t = Pr\left(Z_i = j; x_i, \theta^t\right) = \frac{Pr(X_i = x_i; Z_i = j, \theta^t) Pr(Z_i = j; \theta^t)}{Pr(X_i = i; \theta^t)} \qquad (3.3)$$

Consider the specific case in which the algorithm is applied to a GMM of the form (2.6), where the superscript t is added to indicate the parameters at iteration t, that is, $\theta^t = \left(\pi_1^t, ..., \pi_k^t, \mu_1^t, ..., \mu_{k-1}^t, \Sigma_1^t, ..., \Sigma_k^t\right)$ are the parameters $(\pi_1, ..., \pi_{k-1}, \mu_1, ..., \mu_k, \Sigma_1, ..., \Sigma_k)$ at iteration t. This allows the following (3.4) simplification of the terms in equation (3.3), as well as γ_{ij}^t

itself.

$$
\begin{aligned}
Pr(X_i = x_i; Z_i = j, \theta^t) &= \mathcal{N}(\mu_j^t, \Sigma_j^t)\big|_{x_i} \\
Pr(Z_i = j; \theta^t) &= \pi_j^t \\
Pr(X_i = x_i; \theta^t) &= \sum_{j'=1}^{k} \pi_{j'}^t \, \mathcal{N}(\mu_j^t, \Sigma_j^t)\big|_{x_i} \\
\implies \gamma_{ij}^t &= \frac{\mathcal{N}(\mu_j^t, \Sigma_j^t)\pi_j^t}{\sum_{j'=1}^{k} \pi_{j'}^t \, \mathcal{N}(\mu_j^t, \Sigma_j^t)\Big|_{x_i}}
\end{aligned}
\tag{3.4}
$$

Where $\mathcal{N}(\mu_j^t, \Sigma_j^t)\Big|_{x_i}$ refers to the evaluation of the $\mathcal{N}(\mu_j^t, \Sigma_j^t)$ pdf at x_i. Often θ^0 is obtained by applying the k-means algorithm to cluster the data first and then fitting the k mixture components based off these memberships to obtain θ^0.

Remark. The 'E' in the E-step stands for expectation since it can be viewed as calculating the expectation of an indicator variable. While the 'M' in the M-step stands for maximisation.

When fitting Gaussian mixture models certain simplifications can be made to the E and M steps (McLachlan and Krishnan, 2007). This algorithm is efficiently applied in the Mclust package in R (Scrucca et al., 2016).

Chapter 4

Deciding K

Many algorithms, including the k-means and EM algorithm to the models discussed in chapter 3 require the number of clusters to be specified in advance. In low dimensions graphical methods are sufficient to determine the number of clusters present in data. However visualising data in higher dimensions is problematic and there is a need for methods to determine the number of clusters within data. Additionally, there is a need to distinguish how many clusters are truly present in data, as opposed to ones which are simply due to sampling variation. In this chapter I present five methods (although there are variations of these methods which one may consider as separate) that attempt to find the true number of clusters present in a given set of observations. The five methods are: AIC, BIC, hierarchical SigClust, hierarchical RIFT and sequential RIFT.

4.1 AIC

AIC, Akaike information criterion, can be used to assess any model's quality. It is defined by:

$$\text{AIC} = 2k - 2\hat{\ell},$$

where k denotes the number of independent parameters estimated, and $\hat{\ell}$ is the maximum log-likelihood of the model. AIC aims to select a model by balancing complexity and model fit, and given multiple models the one with the minimum AIC is sought. Notice k constitutes a penalty towards the complexity of the model, while the $\hat{\ell}$ rewards the fit of the model. The penalty towards model complexity is necessary since models with more parameters generally fit data better and yet are more prone to overfitting. A more detailed analysis of AIC can be seen in Bozdogan (1987).

Given observations $X_1, ..., X_n$, let $\hat{p}_i$ denote the Gaussian mixture model fitted to the data with i components. To obtain the preferred k value, first fit K GMM's to the data: $\hat{p}_1, ..., \hat{p}_K$ (assuming a cluster to be Gaussian distribution) and calculate their corresponding AIC scores. The model that achieves the lowest AIC value is the preferred model, and the number of components it has is the number of clusters assumed to be present in the data. The number K is predefined and can be thought of as as upper bound on the number of clusters present.

4.2 BIC

BIC (Schwarz, 1978), Bayesian information criterion, is also a criterion for model selection, BIC is defined as:

$$\text{BIC} = \ln(n)k - 2\hat{\ell},$$

where n is the sample size. k, $\hat{\ell}$ are defined as in AIC. Similarly to AIC, the term $\ln(n)k$ penalises the complexity of the model, while $2\hat{\ell}$ rewards the fit of the model. And hence (again) the minimum BIC is sought. Like AIC, BIC also seeks the optimal trade off between the goodness of fit of the model and the simplicity of the model.

The method to find the number of clusters using BIC is identical to using AIC except BIC has a different penalty term. Notice the penalty for AIC (assuming $n \geq 8$) is less than that for BIC, this causes AIC to pick more complex models.

4.3 SigClust

The SigClust method (Liu et al., 2008; Huang et al., 2015) is more complex than the AIC and BIC methods mentioned above. The base SigClust method tests whether the data has only one cluster or not. This is then extended hierarchically to the Hierarchical SigClust method (SHC) (Kimes et al., 2017) which tests the data for any number (less than the number of data points) of clusters.

4.3.1 Overview

Liu et al. (2008) proposed SigClust, a Monte Carlo method, to evaluate the significance in clustering of data, with applications in HDLSS (high dimension low sample size) settings. Liu et al. (2008) define a cluster to be "data generated from a single multivariate Gaussian distribution". As previously discussed in the distribution based clustering section in chapter 2, this definition is both intuitive and widely used.

The problem is formulated as the hypothesis test:

$$\begin{aligned} H_0 &: \text{ the data are from a single Gaussian distribution} \\ H_1 &: \text{ the data are not from a single Gaussian distribution} \end{aligned} \tag{4.1}$$

Without extension this doesn't allow for testing the significance of $k > 2$ clusters within a dataset. This is explored by Kimes et al. (2017) whereby this hypothesis is tested in a hierarchical fashion. However, to begin let us explore the base case presented in (4.1).

4.3.2 Methodology

Original SigClust Method

Let $\Omega = \{X_1, X_2, ..., X_n\} \in \mathbb{R}^d$ be the observed data. Assuming the null hypothesis (4.1), then the data are from a single Gaussian distribution $\mathcal{N}(\mu, \Sigma)$, $\mu \in \mathbb{R}^d$ is the mean vector, and $\Sigma \in \mathbb{R}^{d \times d}$ is the covariance matrix. Given a clustering of the data: C_1, C_2 the two means clustering index is defined as:

$$CI = \frac{\sum_{X_i \in C_1} \left\| X_i - \bar{C}_1 \right\|^2 + \sum_{X_j \in C_2} \left\| X_j - \bar{C}_2 \right\|^2}{\sum_{X_m \in \Omega} \left\| X_m - \bar{\Omega} \right\|^2} = \frac{ESS_{C_1} + ESS_{C_2}}{ESS_\Omega} \tag{4.2}$$

Liu et al. (2008) claim CI is invariant to location and rotational changes in the data, intuitively this seems true since Euclidean distance only accounts for the distances between points and not their location or orientation in space. Below I show this more rigorously:

- Location: Consider shifting all points in cluster C_1 by arbitrary constant vector v,

$C_1^{Shifted} := \{X_i - v : X_i \in C_1\}$

$$
\begin{aligned}
ESS_{C_1^{Shifted}} &= \sum_{X_i \in C_1^{Shifted}} \left\| X_i - \frac{1}{|C_1^{Shifted}|} \sum_{X_j \in C_1^{Shifted}} X_j \right\|^2 \\
&= \sum_{X_i \in C_1} \left\| (X_i - v) - \frac{1}{|C_1|} \sum_{X_j \in C_1} (X_j - v) \right\|^2 \\
&= \sum_{X_i \in C_1} \left\| X_i - \frac{1}{|C_1|} \sum_{X_j \in C_1} X_j \right\|^2 \\
&= ESS_{C_1}
\end{aligned}
$$

- Rotation: More generally consider the orthogonal transform of a cluster, defined as C_1 multiplied by an arbitrary orthogonal matrix $U^\top$, $C_1^{Rotated} := \{U^\top X_i : X_i \in C_1\}$. Then $ESS_{C_1^{Rotated}}$ can be simplified as below:

$$
\begin{aligned}
ESS_{C_1^{Rotated}} &= \sum_{X_i \in C_1^{Rotated}} \left\| X_i - \frac{1}{|C_1^{Rotated}|} \sum_{X_j \in C_1^{Rotated}} X_j \right\|^2 \\
&= \sum_{X_i \in C_1} \left\| U^\top X_i - \frac{1}{|C_1|} \sum_{X_j \in C_1} U^\top X_j \right\|^2 \\
&= \sum_{X_i \in C_1} \left\| U^\top \left(X_i - \frac{1}{|C_1|} \sum_{X_j \in C_1} X_j \right) \right\|^2 \\
&= \sum_{X_i \in C_1} \left\{ \left(X_i - \frac{1}{|C_1|} \sum_{X_j \in C_1} X_j \right)^\top UU^\top \left(X_i - \frac{1}{|C_1|} \sum_{X_m \in C_1} X_m \right) \right\} \\
&= \sum_{X_i \in C_1} \left\{ \left(X_i - \frac{1}{|C_1|} \sum_{X_j \in C_1} X_j \right)^\top \left(X_i - \frac{1}{|C_1|} \sum_{X_m \in C_1} X_m \right) \right\} \\
&= ESS_{C_1}
\end{aligned}
$$

The invariance of CI to orthogonal transforms and location transforms means that as opposed to working with the null distribution $\mathcal{N}(\mu, \Sigma)$, we can restrict the distribution to be $\mathcal{N}(0, \Lambda)$ where Λ is a diagonal matrix. This allows a significant reduction in the unknown number of parameters of the null distribution from $d + \frac{d(d+1)}{2}$ to d. And therefore it can be seen that this choice of cluster definition (Gaussian) and CI is particularly well suited for HDLSS settings.

For example consider a random sample $X_1, ..., X_n \sim \mathcal{N}(\mu, \Sigma)$ where μ is an arbitrary mean vector and Σ an arbitrary covariance matrix (not necessarily diagonal). Since Σ is real and symmetric, there exists orthogonal U and diagonal Λ such that $\Sigma = U\Lambda U^\top$, where the entries of Λ are the eigenvalues of Σ (this is shown in (Horn and Johnson, 1985)). The transformed sample $U^\top(X_1 - \mu), ..., U^\top(X_n - \mu)$ and $X_1, ..., X_n$ have the same CI. Hence simulating $Y := U^\top(X - \mu) \sim N(0, \Lambda)$ rather than X and calculating their respective CI's give equivalent results, however simulating Y's is preferable since each dimension is separable allowing for more efficient simulation.

Outline

Given a partition of $X_1, ..., X_n$ data points into two clusters: C_1, C_2, SigClust tests the significance of this split by the following steps:

Step 1: Calculate the clustering index CI for the partition: C_1, C_2 by equation (4.2), call this $\tilde{CI}$.

Step 2: Estimate Λ (since CI is invariant under rotation this is assumed to be diagonal).

Step 3: For a specified $N_{sim} \in \mathbb{N}$ times ((Liu et al., 2008) suggest a $N_{sim} \approx 1000$):

(a) Simulate n data points from $\mathcal{N}(0, \Lambda)$ (since CI is invariant under location transform the mean of the distribution is assumed to be 0).

(b) Partition the dataset into two clusters using the k-means algorithm (notice the k-means algorithm with k=2 minimises the CI (4.2)).

(c) Calculate the corresponding clustering index of the partition just calculated.

Step 4: Calculate the empirical distribution of the CI's calculated in step 3, that is the distribution of CI under the null hypothesis. Let the N_{sim} CI's be $CI_1, ...CI_{N_{sim}}$. Then the empirical distribution function is defined as:

$$\hat{F}_{CI}(t) = \frac{1}{N_{sim}} \sum_{i=1}^{N_{sim}} \mathbb{I}_{\{CI_i \leq t\}},$$

where $\mathbb{I}_A$ is the indicator variable of event A.

Step 5: Using this empirical distribution calculate the p-value of the CI of the original dataset (calculated in step 1).

$$p\text{-value} = \hat{F}_{CI}(\tilde{CI})$$

Once the p-value has been obtained it is then trivial to decide whether to reject the hypothesis test at α level.

Note a large N_{sim} value is required to ensure the empirical distribution of the simulated CI's is a good estimate of their true cumulative density function.

Liu et al. (2008) noticed that the empirical distribution of the CI's is approximately Gaussian, and so proposed an alternative method to calculate p-values. This method is to instead fit a Gaussian distribution to $CI_1, ..., CI_{N_{sim}}$. Let this distribution have cdf $\tilde{F}_{CI}$, then p-value= $\tilde{F}_{CI}(\tilde{CI})$. Liu et al. (2008) showed this is an appropriate approximation, and also is preferable, since when there is strong clustering all simulated CI_i's may be larger than all the observed $\tilde{CI}$, and so the empirical p-value is 0. Using the Gaussian approximation allows for a continuous p-value calculation. For example, consider a feasible clustering index c_1 which produces an empirical p-value of 0. Let $0 < c_2 < c_1$, then c_2 also gives an empirical p-value of 0, however when calculating p-values using the Gaussian approximation their p-values would clearly be different, allowing for comparison.

4.3.3 Estimating Λ

There are multiple ways to estimate Λ, as discussed in Liu et al. (2008) and Huang et al. (2015), I briefly outline them below. Note it is assumed Λ is diagonal due to the rotation invariance of the clustering index.

The obvious choice would be to estimate Λ using the eigendecomposition of the sample covariance

matrix, that is:

$$\begin{aligned}\tilde{\Sigma} &= \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})^\top \\ &= U\tilde{\Lambda}U^\top,\ \tilde{\Lambda} := diag(\tilde{\lambda}_1 ..., \tilde{\lambda}_d),\end{aligned} \tag{4.3}$$

where U is orthogonal and $\tilde{\Lambda}$ is diagonal. This decomposition exists since $\tilde{\Lambda}$ is real and symmetric (Horn and Johnson, 1985), and this estimation of Λ is referred to as the sample method. Notice $\text{rank}(\tilde{\Sigma}) = \min\{n-1, d\}$.

In the high dimensional case even with this reduction in parameters to be estimated this method is ill advised, since when $d >> n$ (which is quite common in gene expression data) estimating Λ as in equation (4.3) constitutes estimating d parameters using only n samples. This is undesirable since the true rank of Λ is up to d, however $\text{rank}(\tilde{\Sigma}) = n - 1 << d$, and therefore this estimate is of far lower rank than the true value. It is preferable (Liu et al., 2008) to assume that Λ can be represented by a lower rank matrix and some background noise, this constitutes a factor analysis model.

Factor Analysis Model

Given observations $X_1, X_2, ..., X_n$ of the form:

$$X_i = \begin{bmatrix} X_{i,1} \\ X_{i,2} \\ \vdots \\ X_{i,d} \end{bmatrix}, \tag{4.4}$$

it is assumed that Λ can be represented by a lower rank diagonal matrix $\Lambda^{(0)}$, which constitutes the underlying biology of the sample, and noise ($\sigma_N^2 I_d$). This additional structure means that fewer parameters need to be estimated. The factor analysis structure is:

$$\underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{bmatrix}}_{\Lambda} = \underbrace{\begin{bmatrix} \lambda_1^{(0)} & & & & & \\ & \ddots & & & & \\ & & \lambda_\ell^{(0)} & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix}}_{\Lambda^{(0)}} + \underbrace{\begin{bmatrix} \sigma_N^2 & & \\ & \ddots & \\ & & \sigma_N^2 \end{bmatrix}}_{\sigma_N^2 I_d} \tag{4.5}$$

Estimating $\Lambda^{(0)}$ and σ_N

The standard deviation of a dataset can be robustly estimated using the median absolute deviation from the median, MAD (Rousseeuw and Croux, 1993). Robust in this sense means the estimator performs well even with noise and outliers.

Given n univariate data points $Y_1, ..., Y_n$ their MAD is defined as:

$$MAD = \underset{i=1,...,n}{\text{median}} \left| Y_i - \underset{k=1,...,n}{\text{median}} (Y_k) \right|$$

An estimate for their standard deviation is:

$$\hat{\sigma} = k \cdot MAD$$

where k is a constant depending on the underlying distribution and is needed to make the estimator consistent for the parameter of interest. When assuming Gaussinality and estimating the standard deviation, $k \approx 1.4826$.

Liu et al. (2008) use this method to estimate σ_N. Notice that in generality the data is not necessarily univariate while MAD, as defined in (4.3.3), is defined for only univariate data. To resolve this Liu et al. (2008) use all the $n \times d$ data points to calculate MAD and then use this value to calculate the standard deviation (assuming the underlying distribution is Gaussian). That is, given n observations, the background noise σ_N is estimated as:

$$MAD = \underset{i=1,\dots,n,\ j=1,\dots,d}{\text{median}} \left| X_{i,j} - \underset{k=1,\dots,n,\ l=1,\dots,d}{\text{median}} (X_{k,l}) \right|$$
$$\hat{\sigma}_N = 1.4826 \cdot MAD$$

Now we present two ways to use σ_n to estimate Λ

Hard Thresholding

Liu et al. (2008) suggest estimating Λ, in equation (4.5), to have diagonal entries:

$$\hat{\lambda}_i = \max\{\tilde{\lambda}_i, \hat{\sigma}_N^2\} \text{ for } i = 1, ..., d \tag{4.6}$$

where $\tilde{\lambda}_i$ are the eigenvalues of the sample covariance matrix (as in equation (4.3)). That is, $\tilde{\lambda}_i$'s greater than the estimated background noise σ_N^2 are retained while smaller $\tilde{\lambda}_i$'s are replaced with the noise σ_N^2. In fact Huang et al. (2015) show that when we constrain the rank of $\Lambda^{(0)}$ to be ℓ the maximum likelihood estimate is given by (4.6).

Liu et al. (2008) found that the sample method tends to overestimate the largest eigenvalues of the true covariance matrix, that is, the largest $\tilde{\lambda}_i$ are overestimated, this occurs particularly in the HDLSS setting. To see this consider $d >> n$ then when estimating the sample covariance matrix the total variation of the d dimensions is spread over its $n-1$ non-zero eigenvalues. This is because, as discussed previously, the sample covariance matrix has in this case rank $n-1$, and so $n-1$ non-zero eigenvalues. Then using this hard thresholding method inflates the eigenvalues further as the zero (or close to zero) eigenvalues are replaced by noise, this is discussed in (Baik and Silverstein, 2006). Huang et al. (2015) analyse the properties of hard thresholding method as well as the sample method and found often the sample method was too conservative, while the hard thresholding method was anti-conservative.

Remark. To understand what an anticonservative (and conservative) method is, consider type I and type II errors. Understanding type I and type II errors is best done by considering table 4.1 (James et al., 2021). When considering hypothesis testing it is important to consider the test's

Type I and type II error table		Null hypothesis, H_0, is:	
		True	False
Decision made about null hypothesis, H_0:	Don't reject	Correct	Type II error
	Reject	Type I error	Correct

Table 4.1: Type I and type II error table

error rates. There are two types of error when accepting/rejecting H_0, type I and type II. As shown in table 4.1, type I errors constitute incorrectly rejecting the null hypothesis, while type II errors refer to not rejecting the null hypothesis when it is false.

The level of a hypothesis test specifies an upper bound on the allowable probability of a type I error.

A conservative test is a test with $Pr(\text{type I error})$ less than the level of the test, and hence p-values of a conservative test are usually too large, that is, the test (under H_0) is not rejecting the null hypothesis as often as expected. Note however although this is not desirable a conservative test is still a valid test as it incorrectly rejects the null hypothesis (under H_0) *less* than the test's level.

Conversely an anti-conservative test is a test with $Pr(\text{type I error})$ greater than the level of the test (Sutton, 2004). Therefore it rejects the null hypothesis more than expected (under H_0) and so its p-values are usually smaller than they should be. Anti-conservative tests are not valid tests since they reject the null hypothesis (under H_0) with probability *greater* than the test's level.

Note a test can be conservative in some scenarios while anti-conservative in others (Sutton, 2004).

Also note that type I and type II are often inversely related, that is as one increases the other decreases. To see this consider a test procedure that never rejects the null hypothesis. This clearly has type I error of 0. However it has a significant type II error.

It is also common to use the below notation:

$$\alpha := Pr(\text{type I error})$$
$$\beta := Pr(\text{type II error})$$
$$\text{Power} := 1 - \beta$$

And so the ideal test has small type I error and small type II error (equivalently power near 1).

To attempt to remedy the conservative nature of the test using the sample method, and the anti-conservative nature of the hard thresholding method, Huang et al. (2015) propose another method to estimate the eigenvalues of the covariance matrix- soft thresholding. Note although to motivate the soft thresholding method Huang et al. (2015) provide cases when the sample method is conservative and the hard thresholding method is anti-conservative, they also provide cases when the opposite is true, as well as cases where both methods are conservative or anti-conservative.

Soft Thresholding

(Huang et al., 2015) propose a less aggressive constraint than the hard thresholding method where instead of constraining the rank they constrain the trace of the covariance matrix, Λ. By applying this smoother constraint and performing similar likelihood based analysis as when deriving the hard thresholding estimates, the below estimates for the diagonals of the covariance matrix are found:

$$\hat{\lambda}_i = \max\{\tilde{\lambda}_i - \tau, \hat{\sigma}_N^2\} \text{ for } i = 1, ..., d, \tag{4.7}$$

where τ is a tuning parameter which is linked to the trace constraint. It is clear to see that when $\tau = 0$ the soft thresholding eigenvalue estimates are equal to the hard thresholding estimates and so the hard and soft thresholding methods are identical. Huang et al. (2015) take this to be a lower bound for τ. Huang et al. (2015) also find an upper bound for the value of τ, call this $\tilde{\tau}$[1] and show that when $\tau = \tilde{\tau}$ the soft thresholding and sample methods are identical. Therefore the tuning of τ can be thought of as obtaining a balance between the hard and sample method estimates.

To select the optimal τ Huang et al. (2015) introduce theoretical clustering index (TCI) which, as the name suggests, corresponds to theoretical expected clustering index under the null hypothesis. They further show that it can be written as a function of the eigenvalues of the null distribution's covariance matrix, as in equation (4.8).

$$TCI = 1 - \frac{2}{\pi} \frac{\max_{i=1,..,d} \lambda_i}{\sum_{i=1}^{d} \lambda_i} \tag{4.8}$$

[1] To avoid unnecessary confusion the exact derivation of $\tilde{\tau}$ is omitted, however the derivation can be found in Huang et al. (2015), whereby $\tilde{\tau}$ is defined by the following equation: $\sum_{i=1}^{d} \left(\max\{0, \tilde{\lambda}_i - \tilde{\tau} - \sigma_N^2\} + \sigma_N^2 - \tilde{\lambda}_i\right) = 0$.

Using this notation it can be shown that the optimal $\tau \in [0, \tilde{\tau}]$ is such that it minimises the TCI when TCI is evaluated at the soft thresholding estimates (4.7) (Huang et al., 2015).

4.3.4 Hierarchical SigClust

Kimes et al. (2017) propose a hierarchical testing procedure to test the significance of $k \geq 1$ clusters using methodology similar to SigClust. Broadly speaking it works in a hierarchical fashion, first forming a dendrogram by merging the data in an agglomerative fashion, and then testing select nodes along this dendrogram identifying if the corresponding split is significant.The method proceeds by:

Step 1: Iteratively applying the k-means algorithm (explored in chapter 3) with k=2 to form a dendrogram in an hierarchical (agglomerative) fashion, using Ward's linkage (equation (2.2)) as the linkage criterion and squared Euclidean dissimilarity metric.

Step 2: Beginning at the root (top) node and continuing down through the dendrogram, at each node calculate the significance (p-value) of the split. This is done using the proposed split at that node, and the SigClust method on observations in the corresponding subtree (the subtree of a given node refers to all the nodes and observations connected below that given node).

Step 3: Based on the calculated p-value determine whether splitting at each node is significant. Note we only consider splitting a node if all nodes above it have been deemed significant to split.

At each node j, for $j = 1, ..., n-1$ where node j is the jth node from the top of the dendrogram, (ie. node 1 is the top most node), the following slight variation of the base SigClust hypothesis test (4.1) is performed:

$$\begin{aligned} H_{0j} &: \text{The observations in the subtree of node } j \text{ are from a single Gaussian distribution.} \\ H_{1j} &: \text{The observations in the subtree of node } j \text{ are not from a single Gaussian distribution.} \end{aligned} \tag{4.9}$$

The method above is deliberately vague with regards to how to determine whether or not to split the data at each node. This is because with multiple hypothesis tests to obtain an overall α level test, it is no longer acceptable to reject each test when the p-value is less than α. This is discussed in detail below.

Remark. The p-value at each node need not be calculated, and in fact steps 2 and 3 can be performed simultaneously with calculations and tests omitted for all nodes in the subtrees of nodes for which the null hypothesis is not rejected. Also, at each node a null Gaussian is fitted, this is done using only observations in the corresponding subtree, therefore as the subtree size reduces the Gaussian fit often becomes poorer. To limit this a node is only tested if its subtree contains more than a given number of elements. Kimes et al. (2017) suggest this minimum to be 10.

The choice in Wards linkage as a linkage function is not trivial, Kimes et al. (2017) noticed that in practice it usually gives strong clustering results.

Hierarchical SigClust (SHC) retains the location and rotation invariance inherent to the base Sigclust method and so can be applied in even the HDLSS setting. Additionally, in Step 2 Kimes et al. (2017) suggest using other statistics with similar properties (equivalent to the CI statistic in SigClust), such as using Ward's linkage at each node. Notice Ward's linkage is clearly invariant to location and rotation transforms as it is constructed of ESS values which we showed are location and rotation invariant in chapter 3. Kimes et al. (2017) compared this alternative CI to the standard CI and found that generally this standard CI performed better and so I have continued

using this test statistic (where the standard CI refers to (4.2) and the alternative CI denotes a CI based on Ward's linkage).

Multiple Testing and FWER

Determining whether a node is significant (as in step 3), at α level, is not as simple as rejecting the null hypothesis if the p-value is less than α, where H_0: the data in the node's subtree is from a single Gaussian (and so the node should not be split). This is because multiple tests are being performed testing $H_{01}, H_{02}, ...$ as per equation (4.9).

The probability of incorrectly rejecting a true H_0 significantly inflates along with the number of simultaneously tested hypotheses. To see this consider the case when all H_0 of each individual test (H_{0j}) are considered to be true and independent, and each test is of level α. The probability of rejecting at least one null hypothesis when all these tests are performed is clearly much larger than α. To see this, consider (Fayek, 2021):

$$\begin{aligned}
Pr(\text{Rejecting at least one } H_0) &= 1 - Pr(\text{Not rejecting any } H_{0j}) \\
&= 1 - Pr\left(\bigcap_{i=1}^{r} \text{Not rejecting } H_{0i}\right) \\
&= 1 - \prod_{i=1}^{r} Pr(\text{Not rejecting } H_{0i}) \text{ (by independence of tests)} \\
&= 1 - \prod_{i=1}^{r} (1 - Pr(\text{Rejecting } H_{0i})) \\
&= 1 - \prod_{i=1}^{r} (1 - \alpha) \\
&= 1 - (1 - \alpha)^r
\end{aligned} \tag{4.10}$$

Which (for positive α) tends to 1 as r tends to infinity which is much greater than α (as $1 \geq \alpha$). This is not desirable since as more simultaneous tests are performed (r increases), even when the null hypothesis is true the probability that at least one test incorrectly rejects the null goes to 1. This probability, $Pr(\text{Rejecting at least one } H_0)$, as in equations (4.10), is exactly the family wise error rate ($FWER$). It is the probability, under the null hypothesis of at least one of the individual tests being rejected. It can be thought of as analogous to the type I error, but for a family of multiple tests, more rigorously (James et al., 2021):

$$\begin{aligned}
FWER &:= Pr(\text{Rejecting at least one of the individual test's } H_0) \\
&= Pr\left(\bigcup_{i=1}^{r} A_i\right),
\end{aligned}$$

where A_i is the event of rejecting the null hypothesis of test i. Like the type I error, controlling the $FWER$ is important (Chen et al., 2017). The control of $FWER$ is widely used in multiple testing, particularly for only a few (≤ 100) simultaneous tests, however, as is seen in the equations (4.10) the $FWER$ can become too conservative when the number of simultaneous tests becomes large.

Kimes et al. (2017) seek to control the FEWR of the hierarchical SigClust method at the level α, that is bound it by $\alpha \in (0, 1)$. One such method for doing this is the Bonferoni method.

Bonferoni Method

Recall the Bonferoni inequality (Wasserman, 2006) for any events $A_1, A_2, ...$:

$$Pr\left(\bigcup_i A_i\right) \leq \sum_i Pr(A_i) \tag{4.11}$$

By defining event A_i as the event of rejecting the null hypothesis in test i, then the left hand side of equation (4.11) becomes exactly the $FWER$. Further, let each test have level $\tilde{\alpha}$ (ie. $Pr(A_i) = \tilde{\alpha}$) then equation (4.11) becomes:

$$FWER = Pr\left(\bigcup_{i=1}^{r} A_i\right) \leq \sum_{i=1}^{r} Pr(A_i) = r\tilde{\alpha}$$

And hence to bound $FWER$ by α, selecting $\tilde{\alpha}$ such that $r\tilde{\alpha} = \alpha$ would ensure this criteria is met. This implies that each of the r tests needs to be performed at significance level $\frac{\alpha}{r}$ and therefore the null hypothesis of each test is rejected if it's p-value is less than $\frac{\alpha}{r}$. This is exactly the Bonferoni method.

Despite being commonly used the Bonferroni method is well acknowledged (Chen et al., 2017) to be very conservative particularly when there are a large number of simultaneous hypotheses being tested.

Multiple Testing in SHC

A subtle but important difference between the example demonstrated with equations (4.10) and applying this to the hierarchical SigClust method is that in the later all tests are not performed simultaneously. This is because tests are not performed at nodes which have parent nodes whose null hypothesis is not rejected. One could still use the Bonferroni method at all nodes to produce a valid control of the $FWER$, that is assuming that all $(n-1)$ nodes are being tested and therefore to test each node at significance level $\frac{\alpha}{n-1}$ would give a $FWER \leq \alpha$. This would not only be conservative due to the nature of the Bonferoni method, but also due to the fact that not all $n-1$ nodes are being tested since:

- Nodes with parents nodes that have had their null hypothesis rejected cannot be split.
- As discussed in the method, nodes with less than a given minimum number of data points also cannot be split.

Instead of the Bonferoni method Kimes et al. (2017) correct for multiple testing using a $FWER$ method proposed by Meinshausen (2008), which takes advantage of the hierarchical nature of the SHC method in an attempt to be less conservative than the Bonferoni method.

To bound the $FWER$ by $\alpha \in (0, 1)$ Kimes et al. (2017) perform the multiple tests as follows. Let α_j^* be defined as in equation (4.12)

$$\alpha_j^* := \frac{n_j - 1}{n} \tag{4.12}$$

where n_j is the number of observations clustered at node j.

Hence, beginning at node $j = 1$ and then $j = 2, 3, ...$ to test the hypothesis (4.9) the null hypothesis is rejected if it's p-value is less than α_j^* *and* if the null hypothesis at its parent node was also rejected.

4.4 Relative Information Fit Test (RIFT)

4.4.1 Overview

RIFT (Relative Information Fit Test) (Chakravarti et al., 2019) is similar to SigClust in that it performs a significance test attempting to determine whether data should be split into two clusters, however it has quite different methodology. Like SigClust, it can be extended hierarchically to test the significance of multiple clusters. Additionally, Chakravarti et al. (2019) also propose a non-hierarchical, sequential method to test multiple clusters. Below I discuss the base RIFT method, as well as it's hierarchical and sequential extensions, including theoretical properties that support the method.

Motivation

Chakravarti et al. (2019) showed SigClust is inconsistent (in terms of power) for certain parameters, that is it's power does not tend to 1 as the sample size tends to infinity. More specifically consider n observations of dimension d generated from a mixture of two Gaussians with diagonal covariance matrices:

$$X_i \sim \frac{1}{2}\mathcal{N}(\mu, \Sigma) + \frac{1}{2}\mathcal{N}(-\mu, \Sigma),\ i = 1, ..., n$$

$$\mu = \begin{bmatrix} \frac{a}{2} \\ 0 \\ \vdots \\ 0 \end{bmatrix},\ \Sigma = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{bmatrix},$$

where $0 \neq a \in \mathbb{R}$ is a constant, and $\infty > \lambda_1, \lambda_2 > \lambda_3 \geq \cdots \geq \lambda_d > 0$ Then if the below equation (4.13) is satisfied the SigClust method is inconsistent in terms of power (its power does not tend to 1 as the sample size tends to infinity), while the RIFT method still retains power.

$$\lambda_2 > g(a, \lambda_1), \tag{4.13}$$

where g is a complicated function derived in Chakravarti et al. (2019), but included here for completeness
$g(a, \lambda_1) := \max\left\{\frac{2\lambda_1^2 + \frac{a^4}{16} + \frac{a^2}{2}\sqrt{\lambda_1^2 + \frac{a^4}{64}}}{2\lambda_1}, \frac{\pi}{2}\left(\sqrt{\frac{2\lambda_1}{\pi}}e^{-\frac{a^2}{8\lambda_1}} + \frac{a}{2}Pr\left(|Z| < \frac{a}{2\sqrt{\lambda}}\right)\right)^2\right\}$. Assuming the base SigClust method, SigClust tests the null and alternate hypothesis (4.1), while RIFT tests if the most appropriate two Gaussian mixture model fits the data better than the most appropriate single Gaussian. Importantly it tests the *relative* fit, and hence the method's name (Relative Information Fit Test). This allows it to be more flexible since the true distribution need not be Gaussian or a mixture of two Gaussians, which is preferable from a practical standpoint.

It turns out that developing another statistic is not a trivial task, as often the limiting distributions of the corresponding statistic are intractable (particularly for Gaussian mixture models) as shown in Ghosh and Sen (1984); Liu and Shao (2004).

4.4.2 Methodology

Given observed data $\{X_1, ...X_{2n}\}$ with true distribution p.

Step 1: Split the data randomly into two sets:

- $\mathcal{D}_1$ - for 'training', that is fitting the Gaussian mixture models.
- $\mathcal{D}_2$ - for 'testing', that is performing the hypothesis test.

The split can be arbitrary, however as in Chakravarti et al. (2019) I define the split to be equal ie. without loss of generality $\mathcal{D}_1 = \{X_1, ..., X_n\}$, $\mathcal{D}_2 = \{X_{n+1}, ..., X_{2n}\}$.

Remark. For simplicity I have assumed that the observed data has an even number of points. However, if it does not, the method can still be applied the only difference being the sizes of $\mathcal{D}_1, \mathcal{D}_2$ are no longer exactly equal, and instead the partition can be: $\mathcal{D}_1 = \left\{X_1, ..., X_{\lfloor n \rfloor}\right\}$, $\mathcal{D}_2 = \left\{X_{\lfloor n \rfloor + 1}, ..., X_{2n}\right\}$.

Step 2: Using $\mathcal{D}_1$ fit:

- $\hat{p}_1$ a single Gaussian.
- $\hat{p}_2$ a Gaussian mixture model with two components.

This can be done using the EM algorithm (see chapter 3).

Step 3: Calculate the p-value for the hypothesis test (4.14) and determine whether the null hypothesis is rejected.

$$\begin{aligned} &H_0 : \hat{p}_1 \text{ is closer (or equal) to the true distribution, } p \text{ than } \hat{p}_2. \\ &H_1 : \hat{p}_2 \text{ is closer (or equal) to the true distribution, } p \text{ than } \hat{p}_1. \end{aligned} \tag{4.14}$$

Remark. The hypothesis test (4.14) is formalised below.

Calculating the p-value

In the test statistic to measure the distances between $\hat{p}_1, \hat{p}_2$ and p the Kullback–Leibler distance is considered. Let f, g be two pdf's then the Kullback-Leibler (KL) divergence [2] between f and g is (Wasserman, 2006):

$$\begin{aligned} K(f, g) &\coloneqq \int_{-\infty}^{\infty} f(x) \log\left(\frac{f(x)}{g(x)}\right) \\ &= \mathbb{E}_f\left[\log\left(\frac{f}{g}\right)\right] \end{aligned}$$

The distance $K(f, g)$ can be interpreted as the difference between the distributions f and g, with larger values referring to a larger dissimilarity. $K(f, g) \geq 0$, and $K(f, g) = 0$ if $f = g$ almost surely (Wasserman, 2006). The KL distance can be used to construct a test statistic, to do this consider the difference of KL between $p, \hat{p}_1$ and $p, \hat{p}_2$ as in equation (4.15) below.

$$\begin{aligned} \Gamma &\coloneqq K(p, \hat{p}_1) - K(p, \hat{p}_2) \\ &= \mathbb{E}_p\left[\log\left(\frac{p}{\hat{p}_1}\right)\right] - \mathbb{E}_p\left[\log\left(\frac{p}{\hat{p}_2}\right)\right] \\ &= \mathbb{E}_p\left[\log\left(\frac{p}{\hat{p}_1}\right) - \log\left(\frac{p}{\hat{p}_2}\right)\right] \\ &= \mathbb{E}_p\left[\log p - \log \hat{p}_1 - \log p + \log \hat{p}_2\right] \\ &= \mathbb{E}_p\left[-\log \hat{p}_1 + \log \hat{p}_2\right] \\ &= \mathbb{E}_p\left[\log\left(\frac{\hat{p}_2}{\hat{p}_1}\right)\right] \end{aligned} \tag{4.15}$$

Remark. Although other statistical distances could be used aside from the KL distance, the KL distance, when used as in equation (4.15) has a useful property that p cancels, and only the

[2] The Kullback-Leibler divergence is often called a distance, however this is not strictly correct since it is generally asymmetric, $K(f, g) \neq K(g, f)$.

expectation with respect to p is required. This is incredibly useful since the true distribution is unknown (if it was known, the objective of determining the number of mixture components would become trivial).

Notice that depending on whether Γ is non-negative or not casts two different interpretations:

- $\Gamma \leq 0 \iff K(p, \hat{p}_1) \leq K(p, \hat{p}_2) \iff \hat{p}_1$ is at least as close to the true distribution, p, than $\hat{p}_2$. Where by closer we mean with respect to the KL distance.
- $\Gamma > 0 \iff K(p, \hat{p}_1) > K(p, \hat{p}_2) \iff \hat{p}_2$ is closer to the true distribution, p, than $\hat{p}_1$.

And so a natural hypothesis corresponding to (4.14) would be to test

$$\begin{aligned} &H_0 : \Gamma \leq 0 \\ &H_1 : \Gamma > 0 \qquad \text{conditionally on } \mathcal{D}_1, \end{aligned} \tag{4.16}$$

Remark. Since Γ is a function of $\mathcal{D}_1$ (through $\hat{p}_1, \hat{p}_2$) it is a random variable and so this is a random test.

There are perhaps two intuitive ways to estimate Γ, using a mean based statistic and using the median based statistic. These are discussed below. First, let us define the random variable R, and R_i as it's realisation corresponding to the ith observation, as defined in equation (4.17).

$$\begin{aligned} R &:= \log\left(\frac{\hat{p}_2(X)}{\hat{p}_1(X)}\right), \; X \sim p \\ R_i &:= \log\left(\frac{\hat{p}_2(X_i)}{\hat{p}_1(X_i)}\right) \end{aligned} \tag{4.17}$$

Mean

Since we need to estimate an expectation, $\Gamma = \mathbb{E}[R]$, the obvious choice is to somehow use the sample mean of the R_i's. This is exactly how Chakravarti et al. (2019) proceed in the standard RIFT method. The mean can indeed be used as the data $X_1, ..., X_{2n} \sim p$, and $\hat{p}_1, \hat{p}_2$ are known (they have been fitted in step 2)). Since the test is random, Γ is function of $\mathcal{D}_1$, to perform the test we consider $\mathcal{D}_1$ to be fixed and not random and therefore only data points in $\mathcal{D}_2$ are used to calculate the mean. This allows Γ to be estimated by equation (4.18):

$$\hat{\Gamma} := \frac{1}{n} \sum_{i=n+1}^{2n} R_i \tag{4.18}$$

Chakravarti et al. (2019) show that conditional on $\mathcal{D}_1$, $\frac{\sqrt{n}(\hat{\Gamma}-\Gamma)}{\sigma}$ converges in distribution (by the central limit theorem) to $\mathcal{N}(0,1)$, with $\sigma^2 = Var(R)$. Where a random variable X_n, with cdf F_n is said to converge in distribution to X (with cdf F) if $\lim_{n\to\infty} F_n(t) = F(t)$ for all t for which F is continuous, denoted $F_n \xrightarrow{d} F$ (Wasserman, 2006). $\sigma^2 = Var(R)$ can be estimated by the sample variance $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=n+1}^{2n}\left(R_i - \frac{1}{n}\sum_{i'=n+1}^{2n} R_{i'}\right)^2$. Therefore:

$$\frac{\sqrt{n}\left(\hat{\Gamma} - \Gamma\right)}{\hat{\sigma}} \xrightarrow{d} \mathcal{N}(0,1)$$

Since the null hypothesis is that $\Gamma \leq 0$ given $\hat{\Gamma}$ as calculated by (4.18) p-value can be calculated by performing a one sided test:

$$p\text{-value} = 1 - \Phi^{-1}\left(\frac{\sqrt{n}\hat{\Gamma}}{\hat{\sigma}}\right) \tag{4.19}$$

where Φ^{-1} is the inverse cdf of the standard normal distribution. Once the p-value has been obtained the null hypothesis can be trivially rejected if the p-value is below a given value. Chakravarti et al. (2019) show that when using this test statistic the RIFT method is asymptotically consistent, that is it's power tends to one as the sample size tends to infinity. Hence the lack of power of the SigClust method, seen in equation (4.13) does not occur for the RIFT method when calculating the p-value as in equation (4.19).

Median

Another suitable choice would be to use the median of R_i. This is preferable since the median is a robust estimator and so it is less influenced by extreme values of R_i. Let $\tilde{\Gamma}$ be the population median of R, then consider the hypothesis test (4.20), which follows from the analogous hypothesis test involving the mean (4.16).

$$\begin{aligned} H_0 &: \tilde{\Gamma} \leq 0 \\ H_1 &: \tilde{\Gamma} > 0 \end{aligned} \tag{4.20}$$

To perform this hypothesis test (4.20) the sign test (Conover, 1999) can be used.

The sign test uses the fact that the number of samples above/below the median is binomially distributed, and hence to test the hypothesis (4.20) the median is assumed to be 0 and a one tailed test is performed to see if there are significant number of realisations above 0. This is done by counting the number of values (in our case R_i for $i = n+1, ...2n$) above 0. That is, it assigns a value say R_i^{Ber} dependant on whether $R_i \leq 0$ or $R_i > 0$, as in equation (4.21)

$$R_i^{Ber} = \begin{cases} 1 & \text{if } R_i > 0 \\ 0 & \text{if } R_i \leq 0 \end{cases} \tag{4.21}$$

If the median, $\tilde{\Gamma}$, is 0 one would expect R_i^{Ber} to be the realisation of a Bernoulli trial with probability of success 0.5. And so $\sum_{i=n+1}^{2n} R_i^{Ber}$ is from a binomial distribution with n trials and probability of success 0.5. Therefore if $\sum_{i=n+1}^{2n} R_i^{Ber}$ is significantly larger than np it gives evidence to reject the null hypothesis. More rigorously, let $F_{Y_n}^{-1}$ be the inverse cdf of a binomial distribution with n trials and probability of success 0.5. Then the p-value for the hypothesis (4.20), given our observations is calculated by:

$$p\text{-value} = 1 - F_{Y_n}^{-1}\left(\sum_{i=n+1}^{2n} R_i^{Ber}\right) \tag{4.22}$$

And the hypothesis test can trivially be rejected/not-rejected based off of this p-value. Notice, like before the test is conditional on $\mathcal{D}_1$ hence only data points in $\mathcal{D}_2$ are used to calculate the p value.

Replacing the Kullback-Leibler divergence with the L_2 distance

As mentioned when introducing the Kullback-Leibler divergence, other statistical distance measures exist and can be used. Chakravarti et al. (2019) suggest using the L_2 distance, which for two pdf's f, g is given by:

$$L(f, g) = \int_{-\infty}^{\infty} (f(x) - g(x))^2 \, dx,$$

with $L(f, g) \geq 0$, and a large $L(f, g)$ indicating large dissimilarity between f and g, while $L(f, f) = 0$. Analogous to the method using the KL distance, the difference between $L(p, \hat{p_1})$ and $L(p, \hat{p_1})$ is

calculated and a test statistic is based off of this. Let us calculate $\Theta := L(p, \hat{p_1}) - L(p, \hat{p_2})$:

$$\begin{aligned}
\Theta &= \int_{-\infty}^{\infty} (p(x) - \hat{p}_1(x))^2 \, dx - \int_{-\infty}^{\infty} (p(x) - \hat{p}_2(x))^2 \, dx \\
&= \int_{-\infty}^{\infty} p^2(x) - 2p(x)\hat{p}_1(x) + \hat{p}_1^2(x) - p^2(x) + 2p(x)\hat{p}_2(x) - \hat{p}_2^2(x) dx \\
&= \int_{-\infty}^{\infty} \hat{p}_1^2(x) - \hat{p}_2^2(x) - 2p(x) \left(\hat{p}_1(x) - \hat{p}_2(x)\right) dx \\
&= \int_{-\infty}^{\infty} \hat{p}_1^2(x) dx - \int_{-\infty}^{\infty} \hat{p}_2^2(x) dx - 2 \int_{-\infty}^{\infty} p(x) \left(\hat{p}_1(x) - \hat{p}_2(x)\right) dx
\end{aligned} \tag{4.23}$$

As with Γ, whether Θ is non-negative or not casts two different interpretations,

- $\Theta \leq 0 \iff L(p, \hat{p_1}) \leq L(p, \hat{p_2}) \iff \hat{p}_1$ is at least as close, with respect to the L_2 distance, to the true distribution, p than $\hat{p}_2$.
- $\Theta > 0 \iff L(p, \hat{p_1}) > L(p, \hat{p_2}) \iff \hat{p}_2$ is closer to the true distribution, p than $\hat{p}_1$.

And so a natural hypothesis, conditional on $\mathcal{D}_1$, would be:

$$\begin{aligned}
H_0 &: \Theta \leq 0 \\
H_1 &: \Theta > 0
\end{aligned} \tag{4.24}$$

To simplify calculation of $L(p, \hat{p_1}) - L(p, \hat{p_2})$ consider each of the terms in equation (4.23), $\int_{-\infty}^{\infty} \hat{p}_i^2(x) dx$ for $i = 1, 2$, and $\int_{-\infty}^{\infty} p(x) \left(\hat{p}_1(x) - \hat{p}_2(x)\right) dx$. I will now discuss how each of these integrals can be computed.

- $\int_{-\infty}^{\infty} \hat{p}_i^2(x) dx$ for $i = 1, 2$. Since $\hat{p}_1, \hat{p}_2$ are known, $\int_{-\infty}^{\infty} \hat{p}_1^2(x) dx, \int_{-\infty}^{\infty} \hat{p}_2^2(x) dx$ can theoretically be calculated. Chakravarti et al. (2019) suggest calculating these integrals using importance sampling, which is briefly outlined below.

  Importance sampling (Wasserman, 2006) can be used to integrate $I := \int h(y) dy$ as follows. Let f be a probability density function (that can be sampled from), then

  $$I = \int h(y) dy = \int f(y) \frac{h(y)}{f(y)} dy = \mathbb{E}_f \left[\frac{h(Y)}{f(Y)} \right] \tag{4.25}$$

  And since this is an expectation the mean, $\hat{I}$, serves as an unbiased estimator for I, where $\hat{I}$ is defined in equation (4.26).

  $$\begin{aligned}
  \hat{I} &:= \frac{1}{N} \sum_{i=1}^{N} \frac{h(Y_i)}{f(Y_i)} \\
  \text{where } Y_1, ..., Y_N &\overset{\text{i.i.d}}{\sim} f
  \end{aligned} \tag{4.26}$$

  The full properties of importance sampling can be found in Wasserman (2006). However, broadly speaking, to minimise the variance of $\hat{I}$, f should be similar in shape to h to ensure that there does not exist regions where $\frac{h}{f}$ is large.

  Using notation as in equations (4.25) and (4.26) and applying this method to the integrals $I^{(i)} := \int_{-\infty}^{\infty} \hat{p}_i^2(x) dx$ for $i = 1, 2$, Chakravarti et al. (2019) suggest selecting f to be the pdf of a t distribution with the exact structure at the digression of the reader. In simulations Chakravarti et al. (2019) use $N = 5000$ and assume each dimension of Y_j to be independently t distributed with 1 degree of freedom.

In the case where $h = \hat{p}_i^2$ is not of a similar shape to the t distribution, the estimator, $\hat{I}$, can have large variance. In particular consider the case in which $\hat{p}_i$ has a component with mean of $a >> 0$ in each dimension. When using the t distribution $f((a, ..., a))$ is likely small since the t distribution is centered at 0, where as $\hat{p}_i((a, ..., a))$ (and so $h = \hat{p}_i^2((a, ..., a))$) is likely larger as this is the mean of a cluster, and therefore the ratio $\frac{h}{f}$ is large indicating the method has large variance.

We suggest instead using $f = \hat{p}_i$, since heuristically speaking the shape of this f and h are similar, as f is non negative and h is simply its square. Further, this choice of f accounts for the covariance structure between the dimensions of Y_j, unlike having independent t distributions in each dimension. Additionally, the model $\hat{p}_i$ is known as it has been fitted in an earlier stage of the method, and so simulating from it is feasible (there exists many packages in R, including Rfast (Papadakis et al., 2021) that can efficiently simulate many samples from a GMM), even for large values of N such as $N = 100000$, though this is discussed further in chapter 5.

Let us compare the above two methods for integrating $h = \hat{p}_i^2$, one method, as in Chakravarti et al. (2019), where f is a t distribution in d dimensions with 1 degree of freedom, and the other where $f = \hat{p}_i$. To compare these methods consider the following toy example, a Gaussian mixture model in dimension $d = 1$ with two components, and parameters $\pi_1 = \pi_2 = \frac{1}{2}$, $\Sigma_1 = \Sigma_2 = I_d$ and μ_1, μ_2 with elements that are realisations of independent uniform distributions on the interval $[0, \delta]$. Call this distribution $\hat{p}_2$, let us set $N = 100000$, and calculate the integral of $h = \hat{p}_2^2$ using the above methods and for $\delta \in \{0.1, 100\}$, this is done 200 times and the resulting box plot of the integral values is seen in figure 4.1, with the true integral (calculated numerically) shown in red. Notice in the $\delta = 0.1$ case, where the means

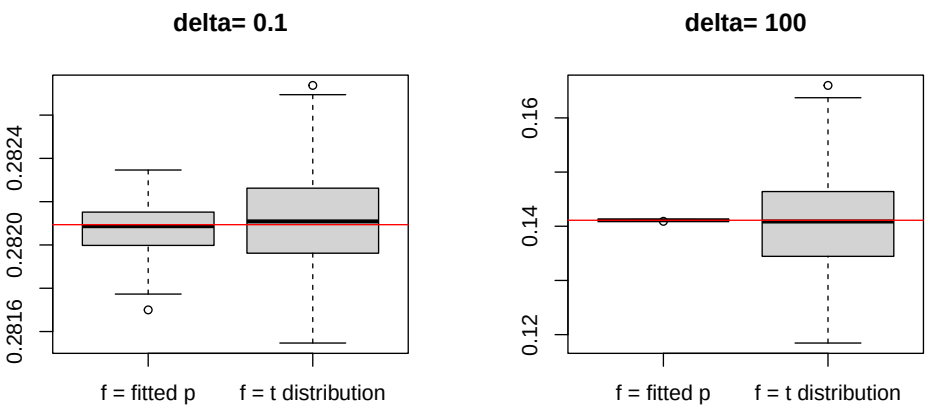


Figure 4.1: Estimates of the integral $\int_{-\infty}^{\infty} \hat{p}_2^2(x)dx$ by importance sampling using different f's, and the true value shown in red. Where $\hat{p}_2$ is a two component Gaussian in dimension $d = 1$ with parameters $\pi_1 = \pi_2 = \frac{1}{2}$, $\Sigma_1 = \Sigma_2 = I_d$ and μ_1, μ_2 each have elements that are realisations of independent uniform distributions on the interval $[0, \delta]$.

of the components of $\hat{p}_2 \approx 0$ both methods perform similarly and obtain a small variance, however the method using $f = \hat{p}_2$ has smaller variance. This trend is only exacerbated when δ increases to $\delta = 100$ (and so the component means are further from 0) with the method using f being t distributed having a large variance such that the estimated integral may be unreliable. Though this example is not rigours it demonstrates the potential issues using f as a t distribution.

- $\int_{-\infty}^{\infty} p(x) \left(\hat{p}_1(x) - \hat{p}_2(x)\right)$

  To integrate this term one can notice that it can be written as an expectation, $\int_{-\infty}^{\infty} p(x) \left(\hat{p}_1(x) - \hat{p}_2(x)\right) = \mathbb{E}_p \left[\hat{p}_1(x) - \hat{p}_2(x)\right]$, this motivates the estimating it by a mean or

median, much like the estimation methods when working with the KL distance.

- Using the mean: $\mathbb{E}_p\left[\hat{p}_1(x) - \hat{p}_2(x)\right] \approx \sum_{i=n+1}^{2n} \hat{p}_1(X_i) - \hat{p}_2(X_i)$
- Using the median: $\mathbb{E}_p\left[\hat{p}_1(x) - \hat{p}_2(x)\right] \approx \text{median}_{i=n+1,\ldots,2n}\left(\hat{p}_1(X_i) - \hat{p}_2(X_i)\right)$

To test the hypothesis (4.24) let us first define the random variable U, and U_i as it's realisation corresponding to the ith observation, this is done in equation (4.27).

$$\begin{aligned} U &:= I^{(1)} - I^{(2)} - 2\left(\hat{p}_1(x) - \hat{p}_2(x)\right),\ X \sim p \\ U_i &:= I^{(1)} - I^{(2)} - 2\left(\hat{p}_1(X_i) - \hat{p}_2(X_i)\right) \end{aligned} \tag{4.27}$$

Using this definition U is analogous to R, and so testing the U's hypothesis (4.24) using U (4.27) is identical to testing the R's hypothesis (4.16) using R (4.17). And therefore the mean and median methods derived can be applied, but now with U_i as opposed to R_i. To summarise the p-values for the mean and median methods using the L_2 distance can be calculated by equations (4.28) and (4.29).

- Mean

$$p\text{-value} = 1 - \Phi^{-1}\left(\frac{\sqrt{n}\hat{\Gamma}}{\hat{\sigma}}\right) \tag{4.28}$$

  Where $\hat{\Theta} := \frac{1}{n}\sum_{i=n+1}^{2n} U_i$ and $\hat{\sigma}^2 := \frac{1}{n}\sum_{i=n+1}^{2n}\left(U_i - \frac{1}{n}\sum_{i'=n+1}^{2n} U_{i'}\right)^2$.

- Median

$$p\text{-value} = 1 - F_{Y_n}^{-1}\left(\sum_{i=n+1}^{2n} U_i^{Ber}\right) \tag{4.29}$$

  Where $F_{Y_n}^{-1}$ is the inverse cdf of the binomial distribution with n trials and probability of success 0.5 and $U_i^{Ber} := \begin{cases} 1 & \text{if } U_i > 0 \\ 0 & \text{if } U_i \leq 0 \end{cases}$.

Therefore once the p-value has been calculated by one of the four methods: Mean (4.19), Median (4.22), Mean L_2 (4.28), Median L_2 (4.29), the hypothesis test can be performed.

4.4.3 Hierarchical RIFT

Let $\mathcal{D}_1, \mathcal{D}_2$ be defined as in the base RIFT method, then the hierarchical method proceeds by:

Step 1: Iteratively fitting mixtures of two Gaussians using $\mathcal{D}_1$ to form a dendrogram in an hierarchical (divisive) fashion. This can be done using the EM algorithm explored in chapter 3. Simultaneously observations in $\mathcal{D}_2$ are partitioned using the mixtures of Gaussians fitted. This can be thought of as a parallel dendrogram containing only observations $\mathcal{D}_2$ and determined by the two Gaussians hierarchically fitted to data in $\mathcal{D}_1$.

Step 2: Beginning at the root (top) node and continuing down through the dendrogram, at each node calculate the significance (p-value) of the split, this can be done using any of the RIFT methods previously discussed (mean, median, mean L_2 or median L_2). The test is performed only on the observations in the subtree of the corresponding node, and is done only on the parallel dendrogram (which includes only observations in $\mathcal{D}_2$).

Step 3: Based on the calculated p-value determine whether splitting at each node is significant. Note only consider splitting a node if all nodes above it have been deemed significant to split.

The similarities between the hierarchical RIFT method and the hierarchical SigClust methods are evident, with the main differences being that in the RIFT method the dendrogram is constructed by fitting GMM's, and a distinction is made between observations used to form the dendrogram and those used to calculate the p-values. Conversely in the SigClust method the dendrogram is formed by the k-means method, and observations are not split for fitting and testing.

As in SHC, steps 2 and 3 can be performed simultaneously and tests omitted for all nodes in the subtrees of nodes for which the null hypothesis is not rejected (split).

Remark. Chakravarti et al. (2019) note that when fitting the mixture of two Gaussians as in step 2 (with the exception of the fit at the root node) fits are performed only on a certain cluster, a certain partition of $\mathbb{R}^d$. Therefore it is more appropriate to fit a truncated Gaussian instead of an un-truncated Gaussian. And hence the usual comparison between the fit of $\hat{p}_1$ and $\hat{p}_2$ should be amended so $\hat{p}_1$ is a truncated Gaussian and $\hat{p}_2$ is a truncated mixture of two Gaussians.

To do this $\hat{p}_i$ should be replaced by $\frac{\hat{p}_i}{P_i(D)}$ where D is the subset of $\mathbb{R}^d$ corresponding to the cluster being tested, and $P_i(D)$ is the probability of Y, $Y \sim \hat{p}_i$ being within this region, D. Chakravarti et al. (2019) suggest estimating $P_i(D)$ by first simulating a sample of size N from $\hat{p}_j$ and then taking the mean number of these data points within the region D. That is, $Y_1, ..., Y_m \sim \hat{p}_j$, $P_i(D) \approx \hat{P}_i(D) = \frac{1}{N}\sum_{j=1}^{N} \mathbb{I}_{\{Y_i \in D\}}$.

One should note that the SHC procedure does not fit truncated Gaussians for testing the null hypothesis. This is a problem since as mentioned above, with the exception of the root node all further nodes are restricted to a certain partition of $\mathbb{R}^d$.

Chakravarti et al. (2019) also propose a hierarchical RIFT method where the dendrogram is tested from the final, bottom, node as opposed to the root node, ie. in a bottom up fashion. The testing procedure then combines clusters until the null hypothesis is rejected. In practice this method was found to perform worse than the analogous top-down method and so is not discussed further in this report.

Controlling $FWER$ in Hierarchical RIFT

To control the family wise error rate at level α Chakravarti et al. (2019) suggest testing node j at level $\tilde{\alpha}_j$ as defined in equation (4.30).

$$\tilde{\alpha}_j := \frac{\alpha}{2^{2\lfloor \log_2(j) \rfloor + 1}} \tag{4.30}$$

To see this does indeed control the error rate at level α. Let there be r hypothesis tests being performed, this r is unknown but less than or equal to $n-1$ since at most $n-1$ hypothesis tests can be performed (one for each node in the dendrogram). Let A_i be the event of incorrectly rejecting the null hypothesis at node i then using the Bonferroni inequality (4.11):

$$\begin{aligned} FWER = Pr\left(\bigcup_{j=1}^{r} A_j\right) &\leq \sum_{j=1}^{r} Pr(A_j) = \sum_{j=1}^{r} \tilde{\alpha}_j \\ = \alpha \sum_{j=1}^{r} \frac{1}{2^{2\lfloor \log_2(j) \rfloor + 1}} &< \alpha \sum_{i=1}^{\infty} \frac{1}{2^{2\lfloor \log_2(j) \rfloor + 1}} = \alpha \end{aligned}$$

Hence when testing each node at level $\tilde{\alpha}_j$ the $FWER$ is indeed controlled at level α. Where the final equality follows from: $\sum_{i=1}^{\infty} \frac{\alpha}{2^{2\lfloor \log_2(j) \rfloor + 1}} = \frac{1}{2} + 2 \cdot \frac{1}{8} + 4 \cdot \frac{1}{32} + 8 \cdot \frac{1}{128} + \cdots = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1$

Remark. Generally controlling the $FWER$ by (4.30) is more conservative than the method suggested by (4.12) since at each node the $\tilde{\alpha}_j$ is smaller than that used by SHC.

4.4.4 Sequential RIFT

Chakravarti et al. (2019) also propose a sequential version of RIFT. Like the hierarchical method it attempts to find the number of clusters in a dataset. However it uses different methodology, which is presented below:

Step 1: Randomly split the data $\{X_1, ..., X_{2n}\}$ into to two approximately equal groups, $\mathcal{D}_1, \mathcal{D}_2$. Without loss of generality let $\mathcal{D}_1 := \{X_1, ..., X_n\}, \mathcal{D}_2 := \{X_{n+1}, ..., X_{2n}\}$

Step 2: For $K := \lfloor \sqrt{n} \rfloor$ fit $\hat{p}_1, ..., \hat{p}_K$ to $\mathcal{D}_1$, where $\hat{p}_j$ is a Gaussian mixture model with j components.

Step 3: For $j = 1, 2, ..., K-1$:

(a) Perform the hypothesis test (4.31).

$$\begin{aligned} H_{0j} &: \Gamma_{js} \leq 0 \text{ for all } s > j \\ H_{1j} &: \Gamma_{js} > 0 \text{ for some } s > j \\ \text{where } \Gamma_{js} &:= K(p, \hat{p}_j) - K(p, \hat{p}_s) \end{aligned} \tag{4.31}$$

(b) Calculate the p-value for for this hypothesis test and reject the null hypothesis if the p-value is below the significance level for the test.

(c) If the null hypothesis H_{0j} is not rejected then use $\hat{p}_j$ to define the clusters, and conclude there are j statistically significant components, otherwise repeat step 3 incrementing j by 1 until $j = K$. If $j = K$ conclude there are at least K clusters present in the data.

The hypothesis (4.31) can seem a little cryptic so consider breaking it down. Notice the null hypothesis can be rephrased as $H_{0j} : \hat{p}_j$ is (at least) closer to the true distribution, p, than $\hat{p}_s$ for all $s > j$, or even more plainly $H_{0j} : \hat{p}_j$ is (at least) closer to the true distribution, p, than all (with K or fewer components) larger GMM's. Here a larger GMM is a GMM consisting of more components.

Remark. K can be arbitrary and, like K in the AIC and BIC methods, is an upper bound for the number of clusters thought to be present in the data.

Testing Null Hypothesis H_{0j}

Consider how to test the hypothesis in step 3a, let R^{js} and R_i^{js} be defined as in equation (4.32), similar to the hierarchical RIFT method's R and R_i.

$$\begin{aligned} R^{js} &:= \log\left(\frac{\hat{p}_s(X)}{\hat{p}_j(X)}\right), \; X \sim p \\ R_i^{js} &:= \log\left(\frac{\hat{p}_s(X_i)}{\hat{p}_j(X_i)}\right) \end{aligned} \tag{4.32}$$

Therefore, like in the hierarchical method, Γ_{js} can be estimated by the mean of R_i^{js}, $\hat{\Gamma}_{js} := \frac{1}{n}\sum_{i=n+1}^{2n} R_i^{js}$. To implement this notice that $\{\Gamma_{js} \leq 0 \text{ for all } s > j\} \iff \{\Gamma_{jj+1}, ..., \Gamma_{jK} \leq 0\} \iff \{\max_{s=j+1,...,K} \Gamma_{js} \leq 0\}$. Hence the null hypothesis can be rewritten as $H_{0j} : \max_{s=j+1,...,K} \Gamma_{js} \leq 0$. By the central limit theorem $\frac{\sqrt{n}(\hat{\Gamma}_{js} - \Gamma_{js})}{\sigma_{js}} \xrightarrow{d} \mathcal{N}(0,1)$, where $\sigma_{js}^2 = Var(\hat{\Gamma})$. And hence the null hypothesis is rejected when the value calculated below (4.33) is less than the desired significance level for the test H_{0j}.

$$1 - \Phi^{-1}\left(\frac{\sqrt{n}\max_{s=j+1,...,K} \hat{\Gamma}_{js}}{\hat{\sigma}_j}\right) \tag{4.33}$$

Let $m > j$ be such that $\hat{\Gamma}_{jm} = \max_{s=j+1,\dots,K} \hat{\Gamma}_{js}$, then

$$\hat{\sigma}_j^2 = \frac{1}{n} \sum_{i=n+1}^{2n} \left(R_i^{jm} - \frac{1}{n} \sum_{i'=n+1}^{2n} R_{i'}^{jm} \right)^2$$

A p-value can also be obtained by considering the median of R^{js}, let $\tilde{R}^{js}$ be the median of R^{js} then performing the sign test would constitute the testing if $\sum_{i=n+1}^{2n} R_i^{Ber,js}$ is $Bin(n, 0.5)$, where $R_i^{Ber,js} := \mathbb{I}_{\{R_i^{js}>0\}}$.

This could be tested for all $s > j$, however like with the introduction of the maximum in the mean method we seek to simplify this test. Notice $\{\tilde{R}^{js} \leq 0 \text{ for all } s > j\} \iff \{\tilde{R}^{jj+1}, \dots, \tilde{R}^{jK} \leq 0\} \iff \{\max_{s=j+1,\dots,K} \tilde{R}^{js} \leq 0\}$, therefore the null hypothesis can be rewritten as $H_{0j} : \max_{s=j+1,\dots,K} \tilde{R}^{js} \leq 0$. With the corresponding test statistic when employing the sign test $\max_{s=j+1,\dots,K} \left\{ \sum_{i=n+1}^{2n} R_i^{Ber,js} \right\}$. Therefore the null hypothesis is rejected when the value calculated below (4.34) is less than the desired significance level for the test H_{0j}.

$$1 - F_{Y_n}^{-1} \left(\max_{s=j+1,\dots,K} \left\{ \sum_{i=n+1}^{2n} R_i^{Ber,js} \right\} \right) \tag{4.34}$$

Where $F_{Y_n}^{-1}$ is the inverse cdf of the $Bin(n, 0.5)$ distribution.

As with the hierarchical method the L_2 distance can be used instead of the Kullback Leibler divergence, and by doing so the hypothesis (4.31) in step 3a, becomes (conditional on $\mathcal{D}_1$) (4.35).

$$\begin{aligned} H_{0j} &: \Theta_{js} \leq 0 \text{ for all } s > j \\ H_{1j} &: \Theta_{js} > 0 \text{ for some } s > j \\ \text{where } \Theta_{js} &:= L(p, \hat{p}_j) - L(p, \hat{p}_s) \end{aligned} \tag{4.35}$$

The remainder of the L_2 method works similarly to the Kullback–Leibler divergence method.

FWER in Sequential RIFT

Chakravarti et al. (2019) suggest testing (4.31) or its corresponding L_2 version at significance level:

$$\alpha_j = \frac{\alpha}{K - j}$$

To see this does indeed control the $FWER$ at level α, one should notice that due to the nature of the method at most only one hypothesis can be rejected as then the procedure terminates. Hence the family wise error rate $FWER = Pr\left(\bigcup_{j=1}^{K-1} A_j\right) \leq \max_{j=1,\dots,K-1} Pr(A_j) = \max_{j=1,\dots,K-1} \alpha_j = \frac{\alpha}{K-(K-1)} = \alpha$, and so controlled at level α.

Chakravarti et al. (2019) show that unlike the AIC or BIC methods the sequential RIFT method offers valid, asymptotic, type I error control.

Chapter 5

Simulations

5.1 Overview

In this chapter we compare the methods discussed in chapter 4 for determining the number of clusters present in data. This is done through a selection of simulations. All code has been implemented in R and can be found on GitHub at `https://github.com/ghicl/m4r`.

Liu et al. (2008); Kimes et al. (2017) have explored the performance of SigClust and its sequential variant in a variety of simulations, as well as demonstrating through simulations that their assumptions hold (including the assumption of normally distributed CI's). With regards to the RIFT method Chakravarti et al. (2019) has also explored the RIFT method in various simulations, including comparison to other methods. However generally simulation studies in the multiple cluster cases are few compared to the base case (testing only if data is from one cluster). And so we aim to perform an extensive simulation study including scenarios not explored before and compare more methods. Additionally we aim to explore differences (if any) in using the modified integration procedure in the L_2 variant (described in chapter 4). Therefore to summarise, in the following simulations we compare sequential RIFT (mean/median and KL/L_2 variants), hierarchical RIFT, AIC, BIC and the hierarchical SigClust methods. The exact form and implementation of each of these methods are discussed below.

5.1.1 Methods

The methods discussed in chapter 4 are analysed in simulation. However many variants and possible adaptations were discussed and so below clarification is given on the variants used, and their implementation is briefly discussed.

AIC and BIC

The AIC method (section 4.1) is used with $K = 14$ for all simulations. This choice of K is such that it allows for over estimation of the number of clusters (which is at most 10 for this simulation study), while not so large to significantly increase computation times.

Analogously to the AIC method the BIC method (section 4.2) is applied with $K = 14$.

Hierarchical SigClust (SHC)

The hierarchical SigClust method (section 4.3.4) uses Euclidean distance as the distance function, Ward's linkage as the linkage and considers a minimum tree size of 10. Additionally, the Gaussian approximation of the CI's (of which 100 are simulated) is used to calculate the p-values, while the covariance matrix is estimated using the soft thresholding technique. These parameters are

selected as they are recommended by Huang et al. (2015) or Kimes et al. (2017). The $FWER$ is controlled at $\alpha = 0.05$. Applying this to the simulations is done using code from Kimes et al. (2017).

Hierarchical RIFT (RHC)

The hierarchical RIFT method (section 4.4.3) which we will refer to as RHC, is applied using the median test statistic based off of the Kullback–Leibler divergence as this is the best performing version (Chakravarti et al., 2019). The top down approach is used with the $FWER$ controlled at $\alpha = 0.05$ using the method discussed in section 4.4.3. This is implemented using the R code from Chakravarti et al. (2019) with slight modifications to allow the code to run on the current R version.

Sequential RIFT (RS)

The sequential RIFT method (section 4.4.4) which we denote RS, is applied with $K = 14$ and $FWER$ controlled at $\alpha = 0.05$. All 4 variants: RS-median and RS-mean (corresponding to the KL distance based methods) and RS-median L_2 and RS-mean L_2 (corresponding to the L_2 distance based methods) are applied.

This method has been coded in R and is available in the GitHub link stated previously. The implementation uses the Mclust package (Scrucca et al., 2016) to efficiently fit Gaussian mixture models. To calculate the corresponding test statistic for the L_2 methods the importance sampling method proposed in section 4.4.4, sampling from $\hat{p}_i$, is implemented with 100000 samples, this is done using the Rfast package (Papadakis et al., 2021) to allow for efficient sampling.

Each of these methods is applied to a sample of $n = 500$ observations with known number of clusters, and its predicted number of clusters recorded. Each sample is drawn from one of the models discussed below. This is repeated 100 times and all results collated.

5.1.2 Simulated Data

In the simulation study, broadly speaking three types of data sets are simulated corresponding to the below models.

Three Clusters

The 'three cluster' set of models consists of a three component mixture model with dimension d from a given distribution with identity covariance matrices and means μ_i for $i = 1, 2, 3$ of the form (5.1).

$$\mu_i = \begin{bmatrix} 1 + (i-1)\delta \\ 2 + (i-1)\delta \\ \vdots \\ d + (i-1)\delta \end{bmatrix} \tag{5.1}$$

Also the weights ($\pi = (\pi_1, \pi_2, \pi_3)$) of the clusters are either even (all weights equal), or uneven (not all weights are equal). Hence the following parameters (5.2) differ within this set of models.

$$\begin{aligned} \text{Distribution} &\in \{t_3 \text{ mixture model}, \text{GMM}\} \\ \delta &\in \{1, 2, 3, 4, 5, 6, 7, 8, 9\} \\ d &\in \{2, 8\} \\ \pi &\in \left\{ \frac{1}{7}(1, 2, 4), \frac{1}{3}(1, 1, 1) \right\} \end{aligned} \tag{5.2}$$

Ten Clusters

The 'ten cluster' set of models is similar to the three cluster models above, with means of the form (5.1) and parameters (5.2). However now the corresponding mixture model consists of ten components and so weights are of the form $\pi \in \left\{\frac{1}{35}(1,2,4,...,4), \frac{1}{10}(1,...,1)\right\}$

Ten Clusters (random)

One issue with the three clusters and ten clusters models is the deterministic nature of the clusters. That is the only variability comes from sampling the models themselves. To add additional stochasticity to the model to ensure that the results are not dependant on the locations of the means we will instead use means of the form (5.3).

$$\mu_i = \begin{bmatrix} U_{i1}^{\delta} \\ U_{i2}^{\delta} \\ \vdots \\ U_{id}^{\delta} \end{bmatrix}, \; U_{ij}^{\delta} \overset{\text{i.i.d}}{\sim} Uniform(0, \delta) \tag{5.3}$$

And the following parameters (5.4).

$$\begin{aligned} \text{Distribution} &\in \{t_3 \text{ mixture model}, \text{GMM}\} \\ \delta &\in \{20, 40, 60, 80, 100, 150, 200\} \\ d &\in \{2, 8\} \\ \pi &\in \left\{\frac{1}{35}(1,1,4,...,4), \frac{1}{10}(1,...,1)\right\} \end{aligned} \tag{5.4}$$

We call this model the 'ten cluster (random)' model. Note, in the R code and the corresponding plots to distinguish the ten cluster models from the ten cluster (random) models in the latter case the distributions are suffixed with '(r)'.

Throughout all these models, δ can be thought of as a control on how close the clusters are to each other, while d is the dimensionality of the data. Also, due to the large number of models simulated only plots displaying interesting trends are included, however all results are saved and available on GitHub, along with all the necessary code to replicate them.

5.2 Simulation Study: Hierarchical Methods

In this section we compare the SHC and RHC methods through a variety of simulations.

5.2.1 Three Clusters

Consider the three clusters models. In the normal distributed clusters case both methods perform well for large δ as seen in figure 5.1. This is seen also in the corresponding uneven cluster model. For small δ the RHC method tends to under-predict the number of clusters while the SHC over-predicts. However this trend is not observed in the t distributed clusters (figure 5.2). In the t distributed clusters both methods over-predict when the clusters are far apart, likely due to the heavy tail nature of the t distribution, however RHC seems to be less affected than SHC.

5.2.2 Ten Clusters

The under-prediction and over-prediction of RHC and SHC methods respectively is again prevalent when considering the ten cluster models (figure 5.3). The RHC method overall performs poorly likely due to the significant number of parameters that need to be estimated to fit a 10 cluster model. Conversely, the SHC method performs well when δ is sufficiently large, with the exception again being when the clusters are t distributed.

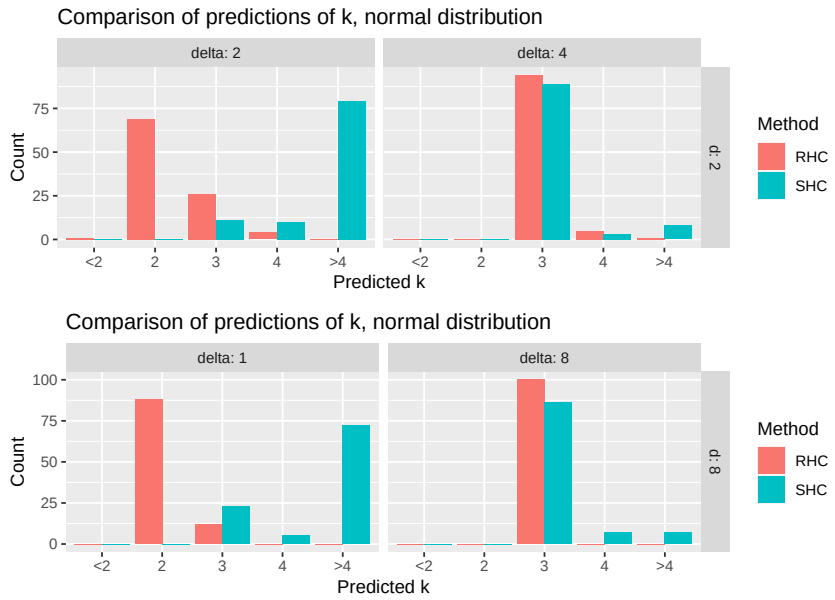


Figure 5.1: Predicted number of clusters by RHC and SHC methods when data is simulated from the three cluster models with even Gaussian clusters.

5.2.3 Ten Clusters (random)

The under/over prediction trends are still prevalent in the ten clusters (random) models except in the 2 dimension t distributed clusters with small δ where the SHC method tends to under-predict (figure 5.4). This is likely due to the fact that in fewer dimensions when selecting the 10 means randomly within a small interval many are very close. This would also explain why RHC performs better in the 10 dimension case. However the SHC method does not. The SHC method still performs poorly with t distributed clusters. Both methods perform almost perfectly for the normally distributed clusters in 8 dimensions (not plotted).

5.2.4 Summary

To summarise there appears to be a trend of RHC under-predicting and SHC over-predicting. As noted by Chakravarti et al. (2019) the number of parameters required to fit the mixture model is a potential disadvantage of the method, as with higher dimensions and/or a larger number of clusters the method performs poorly. However the SHC method tends to lose performance with t distributed clusters while the RHC method is less affected (as noted by Kimes et al. (2017)). Additionally, the uneven cases appear to display no different trends, though overall slightly lower quality predictions.

5.3 Simulation Study: RIFT Methods

In this section we compare the RIFT methods (RHC, RS-mean, RS-mean L_2, RS-median, RS-median L_2) using the models discussed previously.

5.3.1 Three Clusters

For the three cluster model, consider figure 5.5 whereby the methods are compared for both distributions of clusters, with the normally distributed clusters all methods perform comparably, where the RHC method slightly outperforms the RS methods for small δ. The analogous t distributed clusters, also seen in figure 5.5, have overall worse quality predictions even when

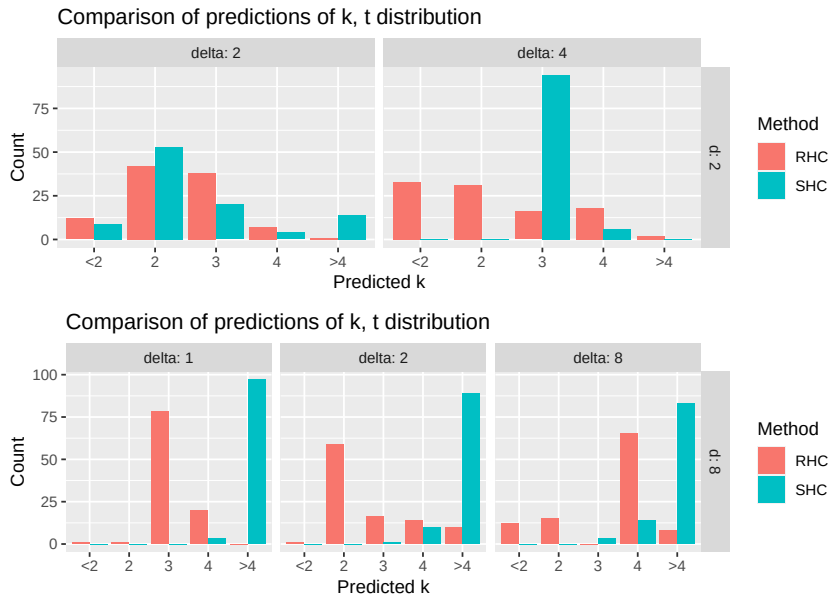


Figure 5.2: Predicted number of clusters by RHC and SHC methods when data is simulated from the three cluster models with even t_3 clusters.

increasing δ (even when $\delta = 9$ the predictions are still poor). Also one can notice that with the t distributed clusters the RS-mean methods tend to under-predict while RS-median methods over-predict. Figure 5.6 compares the even and uneven clusters for $d = 2, \delta = 4$, and shows that with both normal and t distributed clusters when considering uneven clusters the quality of predictions reduces. However the sequential methods perform slightly better than the hierarchical methods.

Notice also that the L_2 variants of the sequential methods perform almost identically to their KL counterparts.

5.3.2 Ten Clusters

Examining the ten cluster models (figure 5.7) we see that the RHC method under-predicts, while the sequential methods perform slightly better, with poor predictions mainly in the small δ cases. As before uneven clusters (figure 5.8) result in an overall decrease in prediction quality. And again for t distributed clusters the RS-mean tends to under-predict while RS-median tends to over-predict. Additionally, the L_2 variants of the sequential methods perform almost identically to their KL counterparts.

5.3.3 Ten Clusters (random)

To examine the ten clusters (random) models consider figure 5.9, in the Gaussian case in 2 dimensions generally under-prediction is seen. Though for larger δ RHC performs better than the sequential variants. Note, when $d = 8$ all methods predict well, hence this figure is not included.

When clusters are t distributed a similar pattern is seen, however in 8 dimensions RHC perhaps performs slightly better than the sequential variants, but no method performs particularly well.

5.3.4 Summary

Overall the sequential methods over predict more than RHC. The tendency for hierarchical RIFT to under-predict may be because of it's conservative $FWER$. However for large enough separation between clusters RHC outperforms the RS methods. For the t distributed clusters the mean methods tend to under-predict while median methods over-predict in the three cluster case and ten cluster case. This is not always true as in the ten cluster (random) case both methods under-predict. While each of the mean and median methods outperform each other in different scenarios overall the median method seems to perform slightly better.

Additionally, the sequential L_2 methods perform almost identically to their KL counterparts throughout all simulations.

5.4 Simulation Study: AIC, BIC, Sequential RIFT

In this section we compare the AIC, BIC and sequential RIFT methods. As established previously, the KL and L_2 methods perform almost identically and so in the below analysis only the KL variants are considered.

5.4.1 Three Clusters

As seen in figure 5.10, for Gaussian clusters when clusters are close together the AIC and BIC methods outperform the RS methods, but for larger δ all methods perform well. However for t distributed clusters all methods perform poorly with perhaps the RS-median performing the best.

The predictions deteriorate in the uneven case, causing slight over-prediction which is more prevalent with the t distributed clusters, as seen in figure 5.11.

5.4.2 Ten Clusters

In the ten cluster models, for Gaussian even clusters at small δ the methods under-predict, however as δ increases their prediction improves. Notice the BIC method tends to require larger δ to predict reasonably - to see this consider figure 5.12 where for $\delta = 3$ with $d = 2$ all methods except BIC are no longer under-predicting. This trend is seen throughout these ten cluster models. Aside from this the methods perform similarly. For t distributed clusters (figure 5.13) it seems the median method is most sensitive to increases in δ as changing this causes its predicted number of clusters to increase (though it still largely over predicts), while the other methods continue to under-predict.

In the uneven cluster case all methods generally perform slightly worse (figure 5.14) however the median method seems to be least effected by the uneven cluster sizes.

5.4.3 Ten Clusters (random)

For the ten clusters (random) models it can be seen, in figure 5.15, that for Gaussian clusters all methods perform similarly, having a tendency to under predict. Except in 8 dimensions they all predict perfectly. With regards to the t distributed clusters when the dimension is increased to 8 the AIC and BIC methods have very few simulations that correctly predict 10 clusters. However the median and mean, for large δ tend to have more correct predictions.

5.4.4 Summary

Overall the AIC and BIC methods perform similarly to the sequential RIFT methods, however when considering uneven clusters the median method seems to perform the best. Additionally in

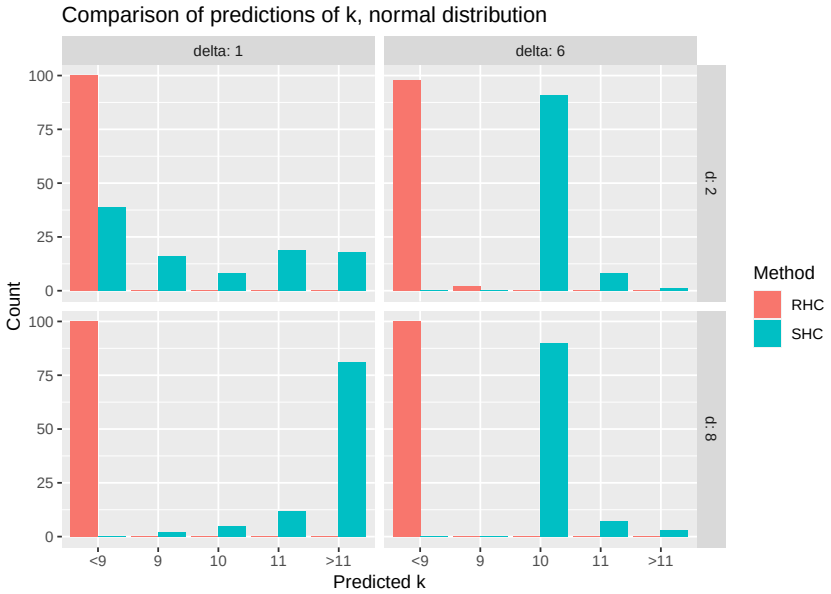
the 10 cluster case the AIC and BIC methods are outperformed by the sequential methods. When considering t distributed clusters the AIC and BIC methods obtain fewer correct predictions (both under and over-predicting) than the sequential methods.

5.5 Summary

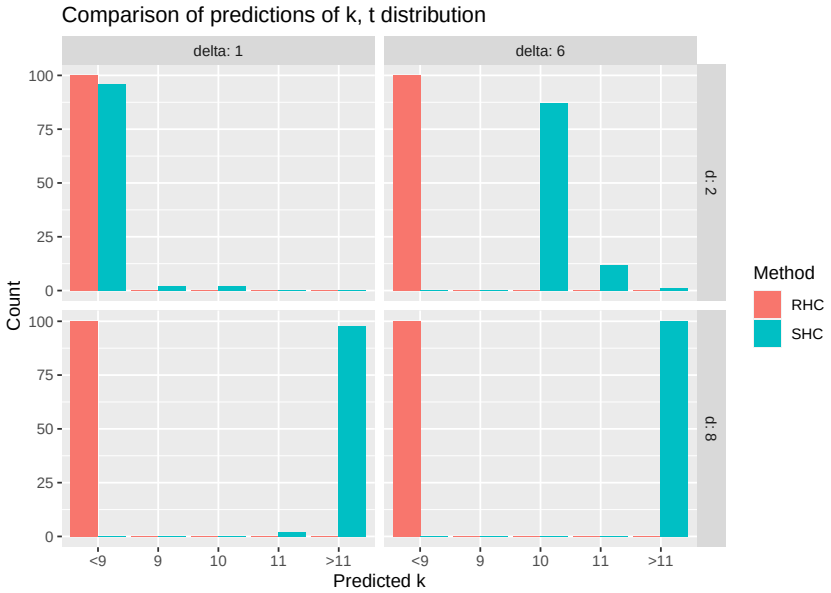
Through an extensive simulation study we observe all methods perform competitively with perhaps the AIC and BIC methods performing the worst. This is expected since a AIC and BIC do not formally test for significant clusters and hence do not have any type I error control.

Perhaps due to the Gaussianity assumption of the SHC method it seems to lack quality when considering non Gaussian clusters, while in the Gaussian case it performs well. Whereas the RIFT methods, which do not assume Gaussianity, predict well for non-Gaussian clusters. Additionally we notice that in the simulations SHC performs poorly it tends to over-predict the number of clusters, while the hierarchical RIFT method tends to under-predict.

The sequential RIFT methods provide good results with the median variant performing best, though generally it is not better than the hierarchical RIFT method. The RIFT methods seem to lose quality as the dimensions and number of clusters being predicted increases, which is also noted by Chakravarti et al. (2019). However the conclusion drawn by Chakravarti et al. (2019) with regards to the differing performance of the L_2 methods is likely due to the different method for integrating the L_2 test statistic. As when integrating using the method we described the L_2 method did not show any difference in performance.

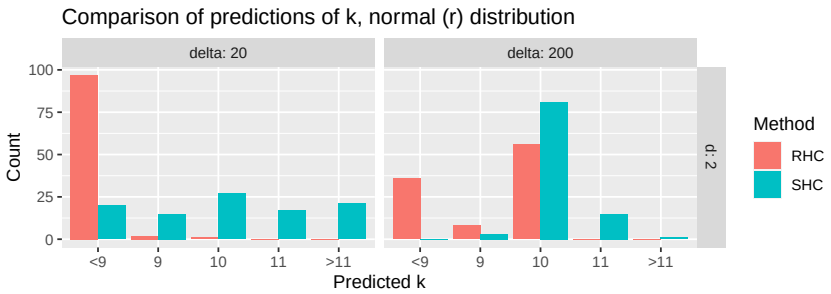


(a) Gaussian clusters

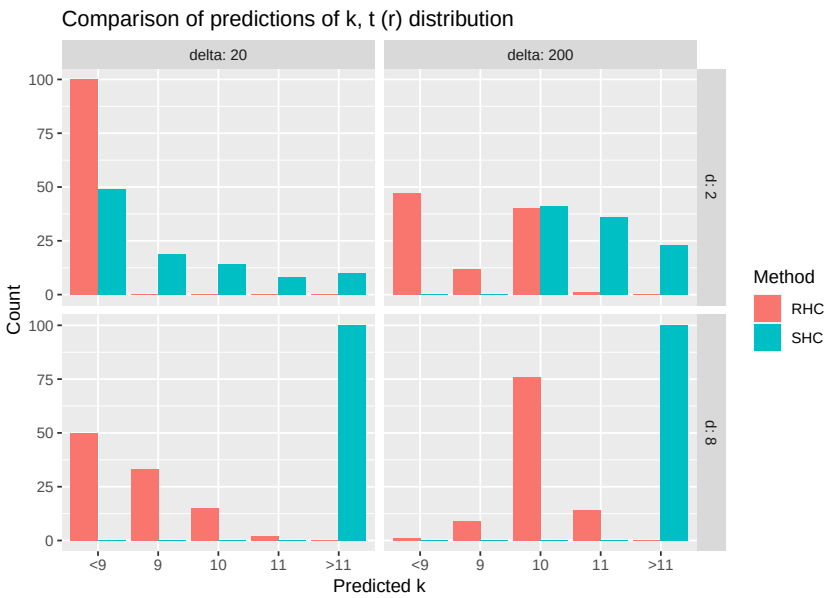


(b) t_3 clusters

Figure 5.3: Predicted number of clusters by RHC and SHC methods when data is simulated from the ten cluster models.

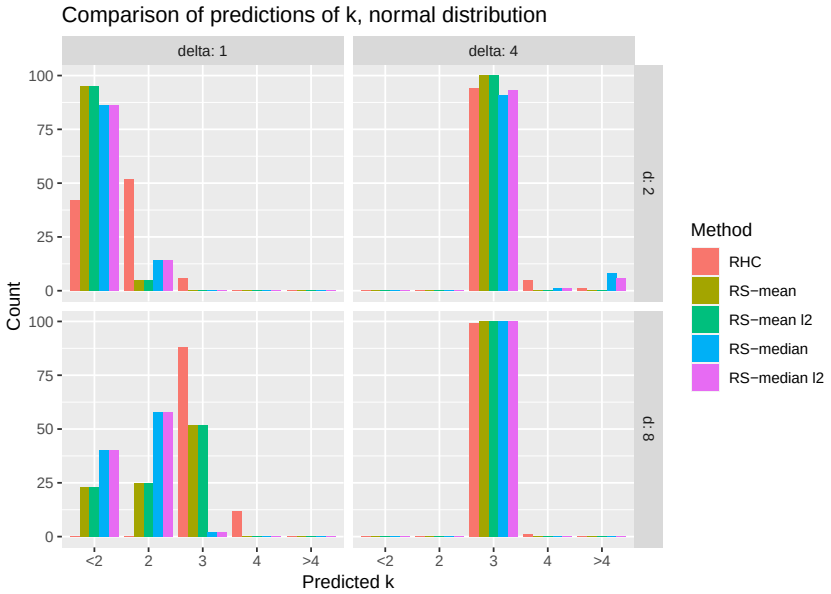


(a) Gaussian clusters

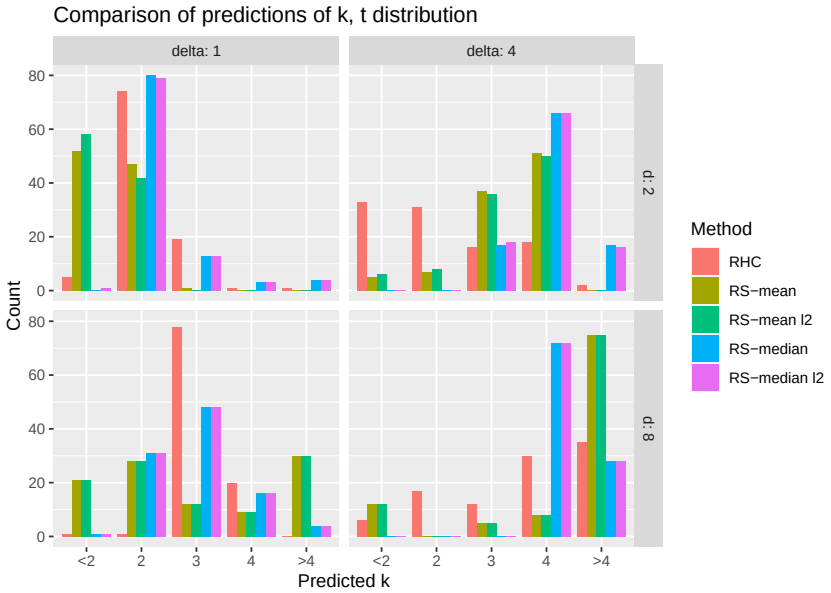


(b) t_3 clusters

Figure 5.4: Predicted number of clusters by RHC and SHC methods when data is simulated from the ten cluster (random) models with even clusters.



(a) Gaussian clusters



(b) t_3 clusters

Figure 5.5: Predicted number of clusters by RIFT methods when data is simulated from the three cluster models with even clusters.

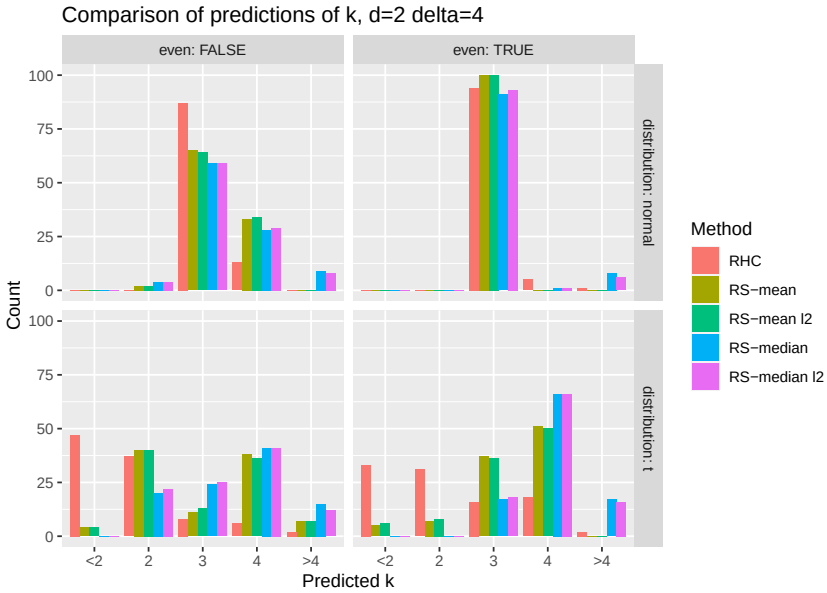
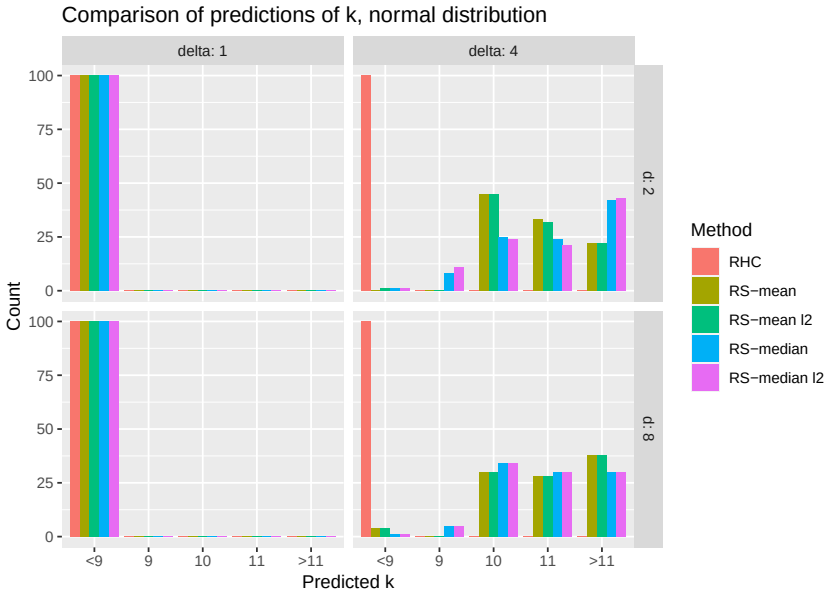
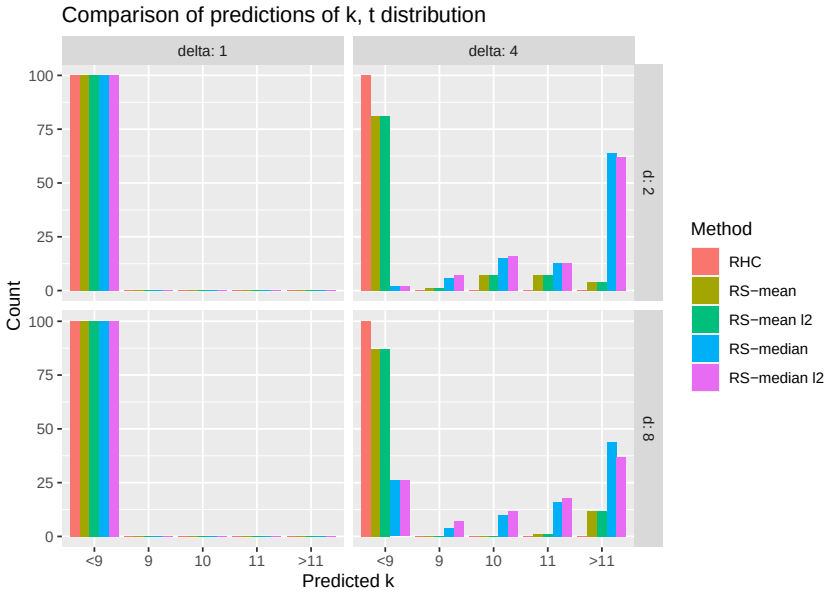


Figure 5.6: Predicted number of clusters by RIFT methods when data is simulated from the three cluster models for $d = 2, \delta = 4$.



(a) Gaussian clusters



(b) t_3 clusters

Figure 5.7: Predicted number of clusters by RIFT methods when data is simulated from the ten cluster models with even clusters.

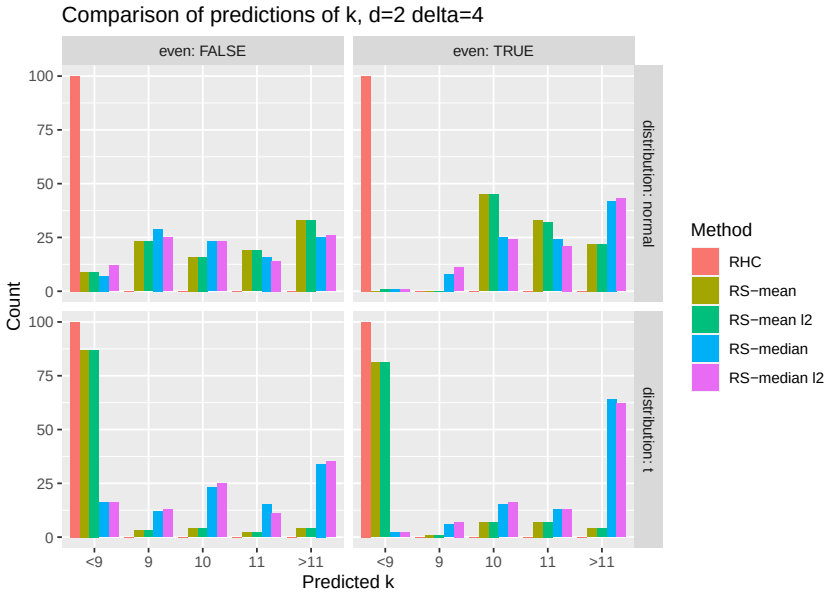
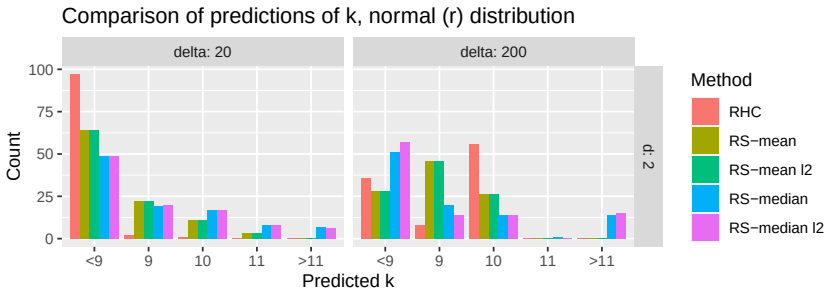
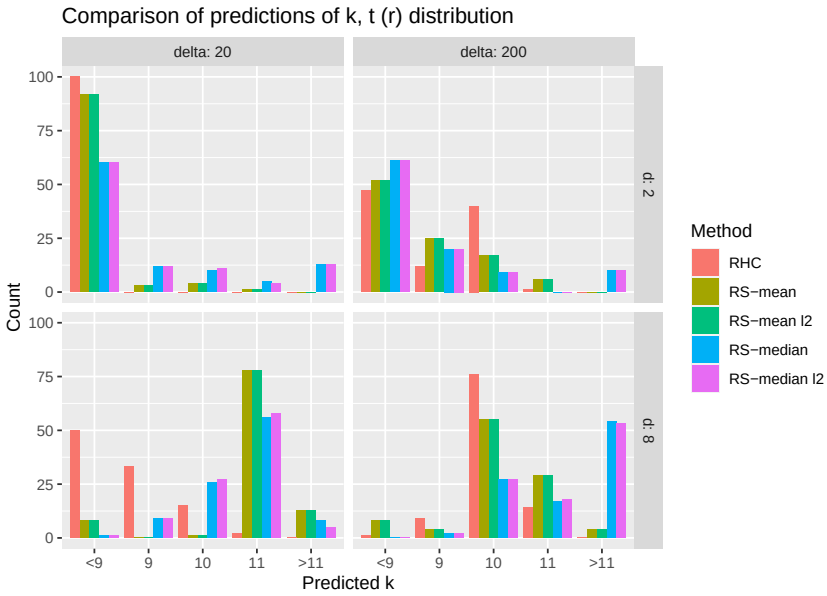


Figure 5.8: Predicted number of clusters by RIFT methods when data is simulated from the ten cluster models for $d = 2, \delta = 4$.

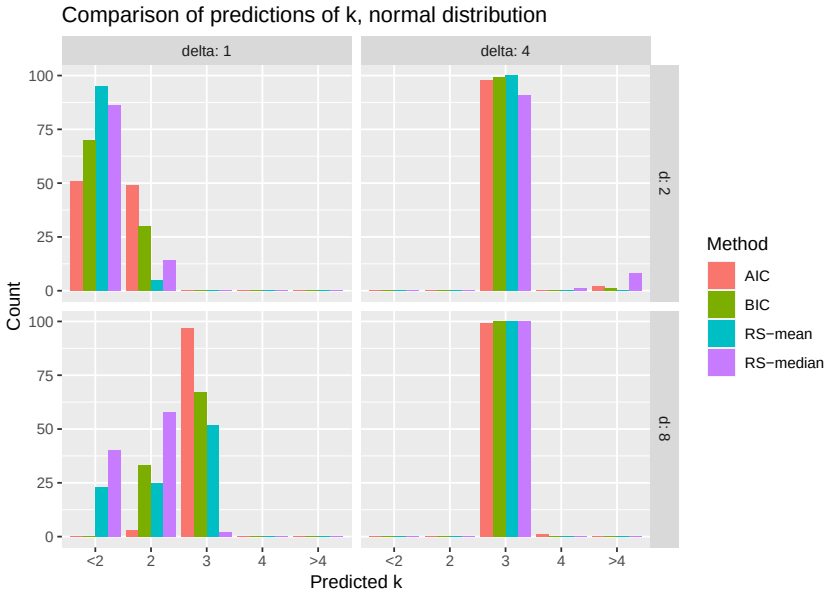


(a) Gaussian clusters

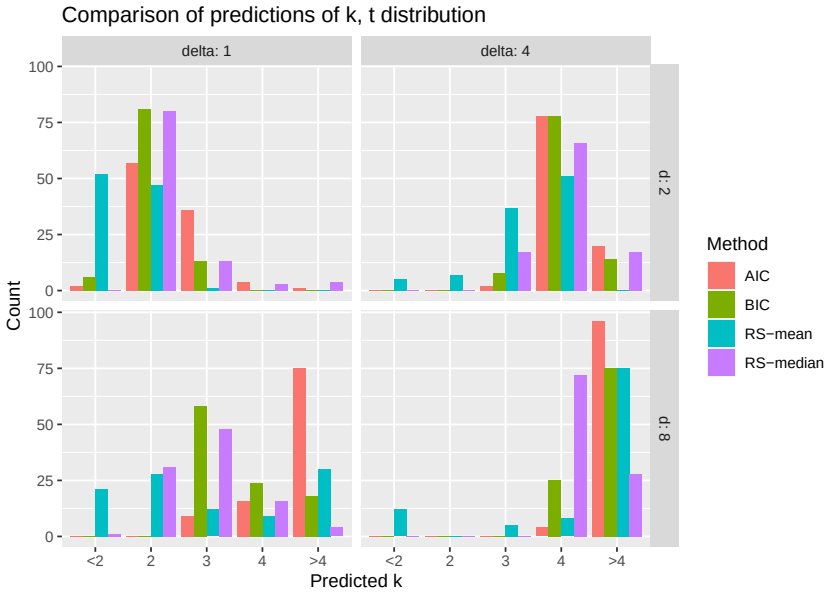


(b) t_3 clusters

Figure 5.9: Predicted number of clusters by RIFT methods when data is simulated from the ten cluster (random) models.



(a) Gaussian clusters



(b) t_3 clusters

Figure 5.10: Predicted number of clusters by AIC, BIC, RS-mean, RS-median when data is simulated from the three cluster models with even clusters.

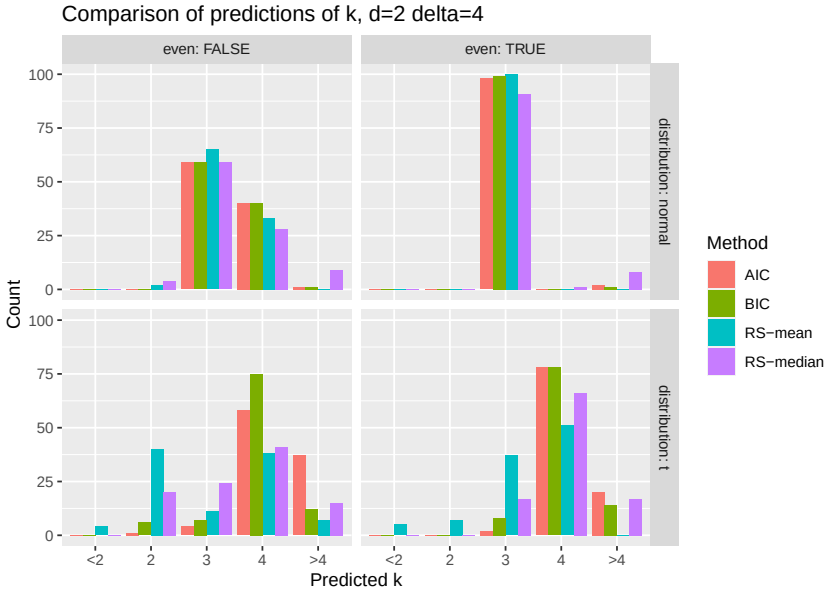


Figure 5.11: Predicted number of clusters by AIC, BIC, RS-mean, RS-median when data is simulated from the three cluster models for $d = 2, \delta = 4$.

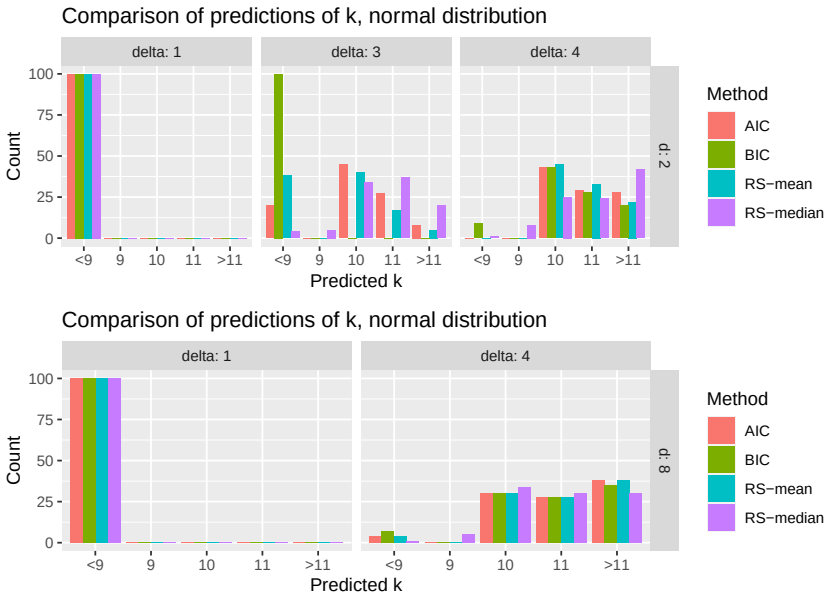


Figure 5.12: Predicted number of clusters by AIC, BIC, RS-mean, RS-median when data is simulated from the ten cluster models with even Gaussian clusters.

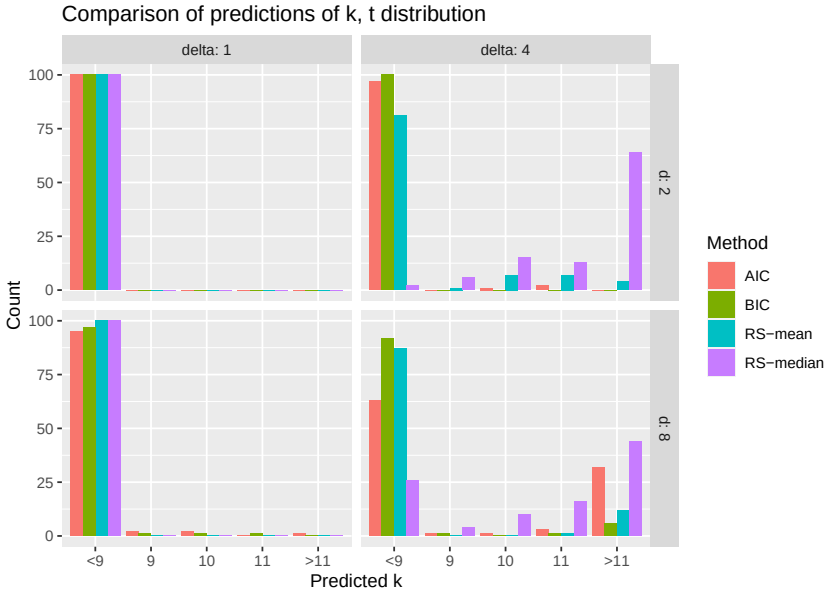


Figure 5.13: Predicted number of clusters by AIC, BIC, RS-mean, RS-median when data is simulated from the ten cluster models with even t_3 clusters.

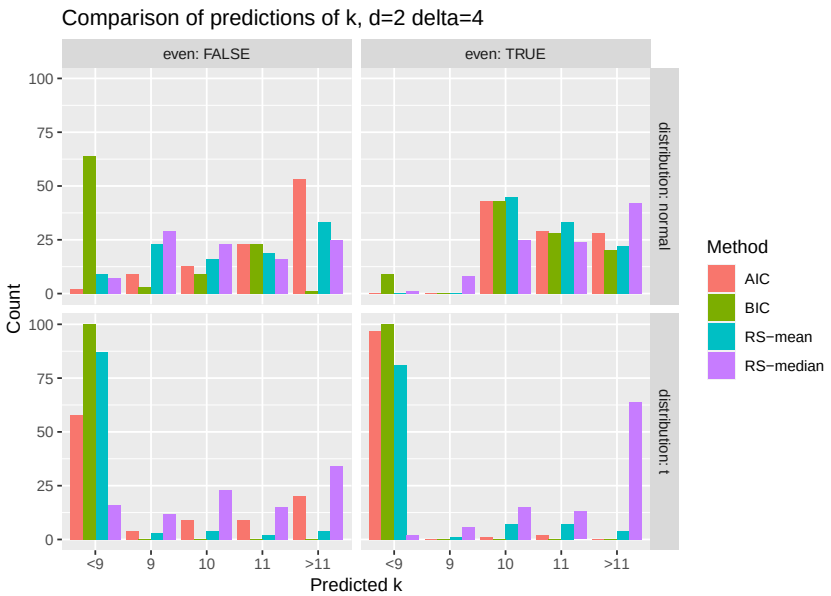
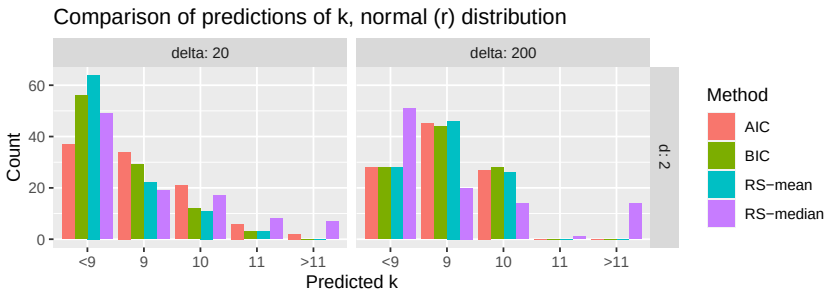
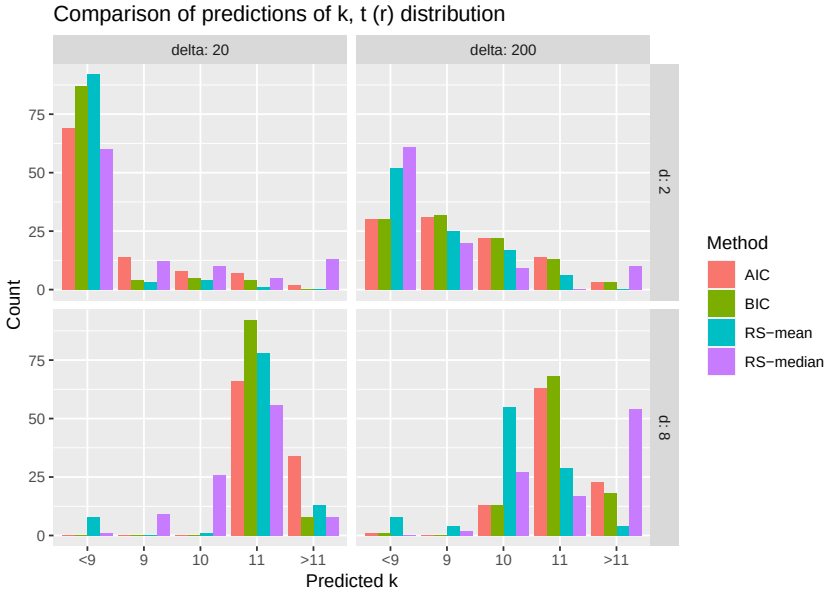


Figure 5.14: Predicted number of clusters by AIC, BIC, RS-mean, RS-median when data is simulated from the ten cluster models for $d = 2, \delta = 4$.



(a) Gaussian clusters



(b) t_3 clusters

Figure 5.15: Predicted number of clusters by AIC, BIC, RS-mean, RS-median when data is simulated from the ten cluster (random) models with even clusters.

Chapter 6

Real Data

To further compare the methods discussed in chapters 4 and 5 we apply them to a cancer gene expression data set. The data set used is from Kimes et al. (2017) and is of the breast cancer gene (BRCA). This is a microarray gene expression data containing 337 BRCA samples. It is compiled and processed as described in Kimes et al. (2017). The data contains 337 samples of 5 molecular subtypes of breast cancer:

- luminal A (97 observations)
- luminal B (54 observations)
- basal-like (91 observations)
- normal breast-like (47 observations)
- HER2-enriched (48 observations)

Due to the high dimensionality of the data (1645 covariates) compared to its number of samples we first reduce its dimensionality by either using principal components analysis (PCA) or by only selecting dimensions that have high median absolute deviation about the median.

6.1 Dimensionality Reduction via PCA

Dimensionality reduction to dimension d can be performed by taking the first d principal components (James et al., 2021). This is done for several d values and then the methods discussed in chapter 5 are applied to predict the number of clusters present. The results are presented in table 6.1, where a $FWER$ of $\alpha = 0.05$ is used.

6.2 Dimensionality Reduction via MAD

The dimensionality of the data can also be reduced using the median absolute deviation about the median (MAD), which is a common technique in gene expression datasets (Chakravarti et al., 2019). To reduce the data to d dimensions one simply takes the d covariates with the largest MAD. This is done for several d values and then methods discussed previously (with a $FWER$ of $\alpha = 0.05$) are applied to determine the number of clusters. The resulting number of predicted clusters is shown in table 6.2.

		d					
		25	50	75	100	125	150
Method	RS-mean	1	1	1	1	1	1
	RS-median	1	1	1	2	1	1
	RS-mean L_2	1	1	1	1	1	1
	RS-median L_2	1	1	1	1	1	1
	BIC	1	1	1	1	1	1
	AIC	2	1	1	1	1	1
	RHC	1	1	1	1	1	1
	SHC	10	3	3	4	4	2

Table 6.1: Table showing predicted number of clusters when applied to BCRA data when using first d principal components.

		d					
		25	50	75	100	125	150
Method	RS-mean	1	1	8	6	10	10
	RS-median	1	1	7	5	2	4
	RS-mean L_2	1	1	8	6	10	10
	RS-median L_2	1	1	7	5	2	4
	BIC	4	2	3	2	2	4
	AIC	7	6	8	6	10	4
	RHC	1	2	5	5	6	1
	SHC	32	23	29	30	27	29

Table 6.2: Table showing predicted number of clusters when applied to BRCA data when using d covariates with the largest MAD.

6.3 Summary

All methods when using dimensionality reduction by PCA perform poorly, with no method correctly predicting 5 clusters. This is probably because PCA is not picking the variables that are responsible for the differences between the cancer subtypes. However when reducing dimensionality using the MAD the RS-median (and RS-median $L2$) and RHC methods are able to correctly identify the number of clusters present in the data when $d = 100$. As the dimensionality, d, increases the RIFT method's prediction quality decreases, likely due to the additional burden of having to fit higher dimensional mixture models.

Chapter 7

Conclusion

7.1 Discussion

To lay the foundation to understand the SHC and RIFT methods this project first explored the SigClust method (Liu et al., 2008) . Following this, we performed an in-depth study of the extension of this method to a hierarchical version (Kimes et al., 2017) which allows the method to determine the number of clusters in a dataset.

Motivated by the lack of power of the hierarchical SigClust method in some scenarios and its strong assumptions we explore in detail the relative information fit test (RIFT) (Chakravarti et al., 2019) which attempts to remedy these issues. We also propose an alternative method for calculating the integrals in the L_2 variant of RIFT.

Once these methods have been explored and implemented, we performed a thorough simulation study on data with greater than 2 clusters (as in this case the hierarchical/sequential variants are needed). In this simulation study we confirm the well established results with regards to the hierarchical SigClust procedure (Kimes et al., 2017; Chakravarti et al., 2019), in that performance deteriorates for non Gausssian clusters. Additionally, we notice that in the simulations where SHC and RHC perform poorly SHC tends to over-predict the number of clusters, while RHC tends to under-predict. We also see that, as noted by Chakravarti et al. (2019), the prediction quality of the RIFT methods deteriorates when the dimension becomes large or there are a large number of true clusters, since in these cases there are a large number of parameters required to fit the GMM's. However the conclusion drawn by Chakravarti et al. (2019) with regards to the differing performance of the L_2 methods is likely due to the different method for integrating the L_2 test statistic, as when integrating using the alternative approach described, the L_2 method performed similarly to the KL version.

Furthermore, we applied the methods to a breast cancer gene dataset and show that the sequential and hierarchical RIFT procedures are able to correctly identify the number of clusters present when using dimensionality reduction by MAD.

7.2 Further Work

In this section we discuss the potential for further work following this project.

We used the SHC method with Euclidean distance and Ward's linkage, however it could be performed with alternative distance measures and linkage criterion, and then the impact of this to prediction quality analysed. Additionally, unlike the hierarchical RIFT method, the hierarchical SigClust method does not fit truncated normals, which as discussed is necessary below the root

node, this could be implemented and the impact studied.

Furthermore, the $FWER$ of the hierarchical RIFT method is controlled in quite a conservative manner, and instead could be controlled using a similar method as in the SHC procedure (from Meinshausen (2008)).

Due to time constraints, the simulation study is performed with only 100 iterations. To improve the reliability of the results found one could increase the simulations and perhaps compare more methods (like Mardia's test (Mardia, 1974) or the gap statistic (Tibshirani et al., 2001)).

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